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Brillouin fiber optic sensors and mobile augmented reality-based digital twins for quantitative safety assessment of underground pipelines

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ABSTRACT

The impossibility of visual inspection and the complexity of combined loads hamper the quantitative assessment, lifetime prediction and control of underground pipelines during their lifecycles. A methodology based on mobile augmented reality (MAR) and Brillouin fiber optic sensors (BFOSs) is presented to build a digital twin (DT) for underground pipelines. Field experiments were carried out to demonstrate that the proposed method can quantitatively assess and predict the structural safety of an underground pipeline from the DT. The results demonstrate that the distributed sensor networks can measure important but unpredictable deformations (i.e., longitudinal bending and axial thermal strain), the joint data-physics driven model can estimate the structural stress state more accurately than the common calculation model, and the MAR-based human–asset interaction interface enables more intuitive, efficient, automated operation and maintenance (O&M). In the future, in-line robotic systems and localized damage models should be further adopted for lifecycle O&M.

1. Introduction

Underground pipelines are widely used engineering assets, and their safety management is easily neglected because they are not visible and direct structural damage inspection is difficult [1]. However, with the rapid growth of underground infrastructure assets and the deterioration of existing structures over time, the increasing number of public emergencies caused by underground pipelines have brought safety management to the forefront [2]. Underground pipelines have a complex spatial distribution and are controlled by different departments. In addition, there are significant differences in the structural states of pipelines caused by specific operating conditions, such as foundation settlement and internal fluid pressure/temperature. Therefore, the digitalization and automation of safety management for underground pipelines is still a challenging issue.

As a next-generation technology that can integrate a series of advanced technologies (e.g., sensors, modeling and visualization), emerging digital twin (DT) technology, which is considered to be an efficient intelligence solution for digitalization and automation, uses digital models to dynamically monitor, describe, simulate, predict and control physical counterparts in a timely manner [3–5]. Recently, DT technology has drawn much attention from both the academic and industrial communities [6]. Gartner predicted that 50% of large industrial corporations would use DT technology by 2021, corresponding to a 10% improvement in effectiveness [7]. Since the National Aeronautics and Space Administration (NASA) gave the first specific definition in 2012 [8], the DT concept has gradually been extended from the field of industrial manufacturing to construction engineering and from theoretical research to application practice [9–13]. At the asset scale, Lin et al. [14] proposed a DT-based collapse fragility assessment method for long-span cable-stayed bridges during strong earthquakes. Mohammadi and Taylor [15] introduced a smart city DT paradigm that revolutionized human–infrastructure interactions and urban planning through the Internet of Things (IoT) and virtual reality (VR). Lu et al. [10,16] presented a system architecture for DT technology that is specifically designed at the building and city levels and developed a DT demonstrator at the West Cambridge site of the University of Cambridge in the UK. Although each of these cases has its own unique definitions and requirements [5], a successful DT paradigm relies on (1) the mutual interaction between the physical asset and its DT, which is related to high-performance sensor technologies and high-fidelity data communication; (2) DT modeling approaches that should focus on both geometrical appearance and the ability to carry out asset-specific analysis, prediction, optimization and decision-making support; and (3) a human–asset interaction interface presented to the end users, which should improve the ability to understand the unique characteristics of an existing physical asset and be

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Table 1
Brief summary of modeling and visualization methods.

References	Key technologies	Key tools/algorithms	Key contribution
Schall et al. (2012) [22]	AR, GIS, and 3D visualization	RTK GPS receiver; tablet PC	Develops a field information system for pipelines, enabling planning, construction and surveys to be executed efficiently on site.
Zhang et al. (2016) [23]	AR, BIM and GIS	A spatial database using the GeoJSON format; CV algorithm; GPS; OpenGL 2.0	Develops and compares two AR solutions (CV version and sensor version) for the BIM and GIS of pipelines.
Soria et al. (2018) [24]	AR, GIS, CAD, 3D visualization	Google Tango; Unity3d engine; ARCore; Mapbox API; Trimble R1	Manages and operates (CRUD) the spatial design information in a way that allows interactions with hidden underground elements.
Napolitano et al. (2018) [25]	VR, SHM data, sensor networks	Kolor Panotour Pro; data integration and visualization	A VR-based connection between SHM data and structures is established in an intuitive and interactive way.
Li et al. (2019) [26]	AR, GIS and database	Unity3d engine; Mapbox API; mobile device	Provides a client–server architecture to develop an AR system that saves time and costs in locating underground pipelines on site.
Zhi et al. (2019) [27]	GIS, Simulation models, 3D visualization	Unity 3d engine; flooding simulation algorithm coupling multisource heterogeneous data	Establishes a 3D dynamic visualization method of an urban drainage appearance model and dynamically simulates a flood disaster.
Fenais et al. (2020) [28,29]	AR, GIS and cloud-based storage	Unity3d engine; ARKit; GPS or GNSS; cloud-based data storage and management platform using the KML format	Enables immersive mapping of underground utilities to guide site construction while avoiding excavation damage to underground pipelines.
Tan et al. (2021) [30,31]	OFDR, Data processing and visualization	DFOS; MATLAB; algorithms for localization, mapping and visualization of SHM data	Develops promising approaches to detect, locate, quantify and visualize pipe corrosion and concrete cracks using a DFOS.

AR = augmented reality; GIS = geographic information system; RTK GPS = real-time kinematic global positioning system; BIM = building information modeling; CV = computer vision; CAD = computer-aided design; CRUD = create-read-update-delete; VR = virtual reality; SHM = structural health monitoring; GNSS = global navigation satellite system; KML = keyhole markup language; OFDR = optical frequency domain reflectometry; and DFOS = distributed fiber optic sensor.

easier and more intuitive to operate.

In pipeline engineering, Siemens proposed the Pipeline 4.0 concept [17]. Pipeline 4.0 utilizes DT technology to explore innovative approaches to further reduce the operating expenses of pipeline systems with no risk or cost. In the current phase, Pipeline 4.0 mainly focuses on developing DTs for pumping stations rather than pipelines. PetroChina has adopted digital recovery techniques, including aerial photogrammetry, 3D laser scanning, 3D terrain construction, oblique photogrammetry and digital 3D modeling, to build the DT of the China–Myanmar oil and gas pipeline [18]. Stennikov et al. [19] presented an object-oriented computer modeling technology to construct the DTs of pipeline systems for online hydraulic calculations. Lieberman et al. [20] discussed the potential benefits of using big data analytics and IoT principles to develop functional DTs of underground infrastructures and improve livability and resilience in modern cities. Carvalho [21] presented a DT framework combining building information modeling (BIM) and asset management (AM) techniques to support the maintenance and operation of water and wastewater utility assets. According to a literature survey, pipeline DT technology is still in its infancy, and there are few reports on using DTs for the structural safety assessment of underground pipelines.

The application of DTs to underground pipelines faces the following challenges. (1) The sensor network technique underpins the mutual interactions between DTs and hidden buried assets. Although nondestructive test (NDT)-based in-line inspection (ILI) tools, including magnetic flux leakage (MFL), ultrasonic wall measurement (UT) and remote field eddy current (RFEC) tools, are commonly adopted in oil and gas pipelines, they are local inspection techniques rather than real-time sensing techniques. However, these methods cannot measure the stress states along a pipeline. The supervisory control and data acquisition (SCADA) system uses only sparse low-sampling transducers to monitor the internal pressure, flow rate and temperature parameters of pipelines; however, this system is unable to provide information regarding structural deterioration during a lifecycle. The comprehensive sensing of the structural state, including both stress/deformation and deterioration in the pipeline, is recognized as one of the most critical issues for pipeline DTs. (2) Human–asset interactions and visualization are vital for underground pipelines since they are not visible in the buried environment. The state of the art is summarized in Table 1. Augmented reality (AR) is considered to be a friendly human–asset interaction technique due to its fast communication, 3D visualization and beyond-reality interaction for underground pipelines. However, the current AR-based modeling and visualization methods mainly focus on the geometric attribute information (design and construction phases) rather than the dynamic evolution of structural states (operation and maintenance (O&M) phase). During the O&M phase, which covers >80% of the lifecycle, pipelines suffer from corrosion and load or operating condition changes, resulting in the dynamic evolution and deterioration of the structural state over time. Therefore, the current studies lack a modeling and interaction approach for the dynamic evolution of the pipeline structural state to guide lifecycle safety management.

To address these challenges, a framework for DTs is presented in this study for underground pipelines by using mobile augmented reality (MAR) and Brillouin fiber optic sensors (BFOSs). BFOSs, in addition to IoT-based pressure and temperature sensors, constitute distributed sensor networks, which are used to comprehensively sense the structural stress states of underground pipelines. Then, a joint data-physics driven model is presented to achieve dynamic modeling of the evolution and deterioration of specific asset structures. A MAR-based human–asset interaction interface is proposed to visualize geometric attributes, real-time monitoring data and structural safety assessment results. The feasibility of the proposed methodology for automated safety assessment was validated by experimental investigations in a real pipe network.

2. Methodology development

Underground pipelines are not visible and are large-scale. These pipelines are subjected to complex combined loads and inevitably suffer from structural deterioration during long-term service. A framework of a DT system based on MAR and BFOSs is proposed for quantitative safety assessment and O&M decision-making support. As shown in Fig. 1, the DT of an underground pipeline has three main components: distributed sensor networks oriented toward the physical pipeline, a joint data–physics driven model oriented toward the DT, and a human–asset interaction interface based on MAR oriented toward safety inspectors. Based on this framework, the methodology is presented in detail as follows. It should be noted that only continuous metallic pipelines are investigated in this study, and jointed pipelines (e.g., bell-and-spigot cast-iron and concrete pipes) are beyond the scope of the proposed approach.

2.1. Distributed sensor networks

The sensing scheme for DTs depends on the structural failure modes and deterioration mechanism. In general, a combination of circumstances leads to pipeline failure. Pipeline breakage is likely to occur when operational and environmental loads are imposed on deteriorated pipes with corrosion, degradation and distress. According to O'Day et al.

[32] and Rajani and Klenier [33], the main failure modes of pipelines can be categorized into circumferential and longitudinal breaks. Circumferential breaks are mainly caused by longitudinal stress due to thermal contraction, soil settlement or voids, and third-party interference. Longitudinal breaks are the result of excessive hoop stress caused by internal pressure, soil cover, frost load, structural deterioration, etc. Current design codes clearly present a calculation model for hoop stress from external loads (soil overburden and traffic load) and internal pressure [34,35]. The thermal axial stress in a continuous pipeline is also considered in the model. Rajani et al. [36] developed a pipe–soil interaction analytical model for the longitudinal stress of jointed pipelines. This model accounts for temperature differentials and explains the frequently observed occurrence of circumferential breaks in pipelines. Notably, the longitudinal bending stress due to foundation defects (settlements and voids) or third-party interference is not considered in the model because these events cannot be predicted in advance. During long-term service, the longitudinal bending actions induced by foundation defects and third-party interference may occur with unforeseen magnitudes and at arbitrary times and locations during the lifecycle. However, third-party interference and foundation defects are the second and third greatest contributions to pipeline failures [37–39], respectively.

As indicated earlier, longitudinal bending stress may occur arbitrarily along pipelines. The commonly used discrete point sensors, i.e., strain gauges and fiber Bragg grating (FBG) sensors, can only measure the structural strain/stress in limited specific locations. This contradicts the fact that the locations of the longitudinal actions on a pipeline cannot be determined in advance. As a cutting-edge sensing technique, Brillouin backscattering-based fiber optic sensors can provide real distributed measurements for strain as well as temperature with a sensing range of up to a few tens of kilometers [40–43]. BFOSs are ideal candidates for monitoring pipelines due to their large spatial range and relatively small spatial resolution [44]. Inaudi and Glisic [45] reported several significant field application examples using BFOSs, including leakage detection for brine and gas pipelines and strain and deformation monitoring for gas pipelines. Based on strain and temperature monitoring data from BFOSs and structural analysis of buried pipelines, Feng et al. [46] proposed a methodology for pipeline structural performance assessment and applied it to practical cases. The capability of distributed sensors to accurately measure the strain/stress state relies on their positions on the pipeline. To monitor the longitudinal strain, mounting a single Brillouin fiber optic strain sensor on the pipe crown was proposed, as shown in Fig. 1. Ni et al. [47] and Lan et al. [48] showed that the flexural response (i.e., bending deformation) is dominant in the longitudinal strain caused by surface faulting. With the exception of geohazards and foundation defects, according to Li and Feng [49], the failure of oil and gas pipelines and heating pipelines can be caused by excessive thermal stress. The thermal exchange process between the internal hot fluid and the pipe commonly lasts only a few hours before reaching a steady state; thus, we assume that thermal loading does not occur simultaneously with geohazards and foundation defects. If no thermal action is identified from the monitoring data, the measurements of a single BFOS will only reflect the bending responses in the vertical plane. The proposed single-sensor configuration has the advantage of easy installation. At the same time, a distributed fiber optic temperature sensor is placed in parallel along the strain sensor on the pipe crown. This sensor independently measures the temperature around the pipeline, so it can provide leakage detection [50] as well as thermal compensation for the strain sensor. The proposed sensing scheme provides distributed real-time monitoring of the longitudinal strain/stress for a pipeline DT.

Hoop stress in a pipeline is created by the soil cover, traffic load, and internal pressure. Even with BFOSs, distributed monitoring of hoop stress is not cost-effective or practical for underground pipelines. In previous research [34,36,46,51–53], theoretical models were provided for hoop stress in underground metallic pipelines with diameters <500 mm. These theoretical models are related to critical loads, such as internal pressure, overburden loads and traffic loads. According to Ahammed and Melcher [34] and Mahmoodian and Li [51], the variations in surface loads (soil overburden and traffic load) have low contributions to failure with increased exposure period. Attention should be given to abnormal events related to surface loads by routine inspection instead of real-time monitoring. The internal pressure, temperature differential, and continuous corrosion play significant roles in pipeline failures [34,54]. Thus, we propose a sensing scheme for the real-time monitoring of internal pressure and temperature. The IoT-based digital temperature sensor and pressure transmitter are placed at a certain distance along the pipeline. By using the measured internal pressure and temperature data, the hoop stress can be accurately estimated by the well-developed analytical model [34]. The effect of corrosion was also considered in the stress analysis by introducing the corrosion model [51]. With increasing elapsed service time, well-planned inspections should be conducted periodically, and the data from ILI tools can be utilized to calibrate and update the corrosion model. In conjunction with BFOSs, digital pressure and temperature gauges are used in real-time sensor networks for the pipeline DT.

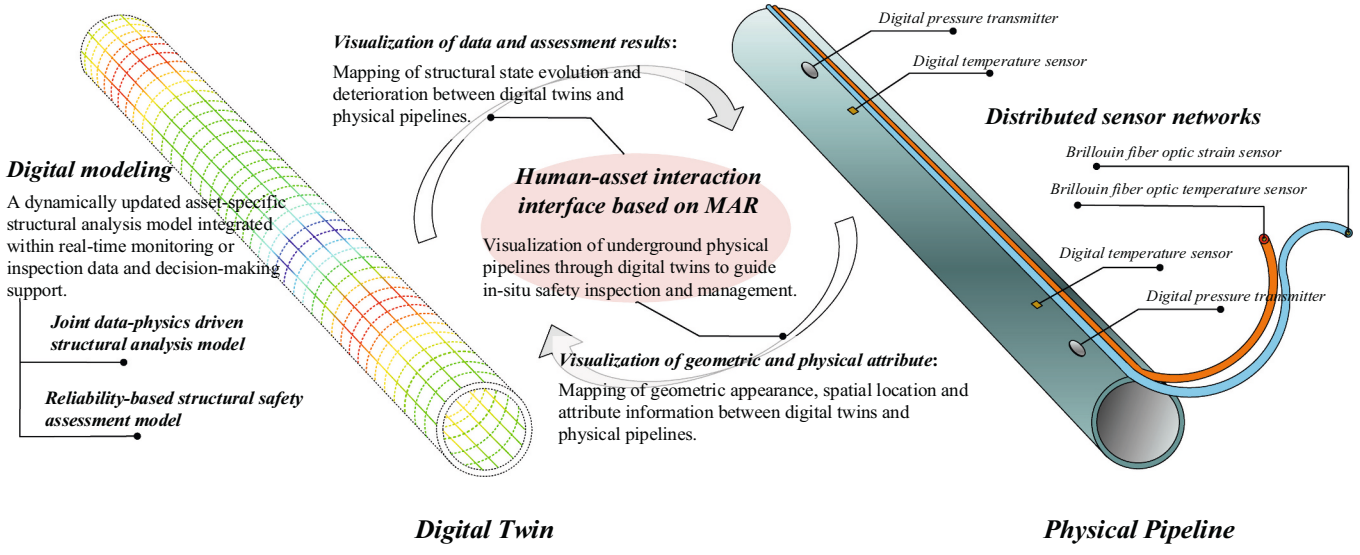


Fig. 1. Framework of the DT for an underground pipeline.

2.2. Digital modeling

A DT is a computational model or a set of coupled computation models that evolves over time to persistently represent the structure, behavior and context of a unique physical asset [55,56]. To fulfill this goal, different models or model types can be adopted for different purposes in DTs. In the construction sector, BIM or geographic information system (GIS)-based modeling approaches are adopted to establish visual models at the building and city levels [57,58]. In addition to geometry visualization, data-driven modeling approaches have mostly been used to map physical assets into a DT [16]. Smart data analysis using machine learning (ML) and artificial intelligence (AI) play a major role in the data-driven model [59]. However, the data-driven model has some downsides. The most serious limitation is the lack of interpretability. In particular, for safety-critical applications, black-box approaches are unacceptable. Physics-based modeling is the classic approach to analyzing, simulating and predicting the physical world. The major advantage of physics-based approaches is that they are generally less biased than data-driven models because they are mathematical descriptions that characterize physical phenomena [60]. With the rapid development of high-performance simulators, physics-based modeling approaches have been used to establish DT technology. The disadvantage of physics-based models is that they rely on parametric assumptions. Therefore, they are mainly limited to the design phase of engineering systems.

The overarching goal of this study is to develop digital models for data analysis, safety assessment, and health prediction of pipelines. As presented in Section 2.1, the actual longitudinal bending behavior in a pipeline cannot be determined by physics-based models in an a priori manner. On the other hand, hoop stress should be estimated by using a structural analysis model with monitoring data (internal pressure and temperature). Neither pure physics-based nor pure data-driven approaches can individually model the stress state of pipelines. To close this gap, a joint data-physics-driven model is proposed for data analysis and structural state assessment in this study. This novel approach is intended to overcome the limitations of pure physics-based and pure data-driven modeling methods. This model introduces monitoring data, i.e., longitudinal strain, internal pressure, and temperature differential data, into a well-understood physics-based model to accurately obtain structural stress in a real-time manner. By utilizing real data, the new modeling approach eliminates the shortfalls of parametric assumptions and simultaneously improves the interpretability, accuracy and robustness of the model. Then, a reliability-based approach incorporating the joint data-physics-based structural analysis model is proposed for online structural safety assessment. The presented modeling methodology allows DTs to dynamically update computational models integrated with observational data so that they reflect the changing structural state of physical pipelines.

2.2.1. Joint data-physics-driven structural analysis model

It is well known that the stress state is a direct indicator of structural safety. In fact, pipe–soil interactions are complex under combined loads, which leads to a 3D stress state in the pipeline. The stress at an arbitrary location of the pipe wall includes three components, i.e., longitudinal stress σ_l, hoop stress σ_h, and radial stress σ_r. According to references [34, 61, 62], the radial stress can be neglected in the Tresca or Von Mises failure criteria because the maximum principal stress mainly depends on hoop and longitudinal stress for underground pipelines (Fig. 2). In general, the effects of internal pressure, temperature differential, and bending actions on the longitudinal stress σ_l are given as follows:

$$\sigma_l(x) = \sigma_l^p(x) + \sigma_l^T(x) + \sigma_l^b(x) \tag{1}$$

where x is an arbitrary location along the pipeline and σ_l^p, σ_l^T, and σ_l^b are the stress components corresponding to internal pressure, temperature differential and longitudinal bending, respectively. σ_l^p is created by the Poisson effect of hoop deformation under internal pressure.

Similarly, the hoop stress σ_h is a result of internal pressure (σ_h^p), soil pressure (σ_h^s), and traffic load (σ_h^V) as follows:

$$\sigma_h(x) = \sigma_h^p(x) + \sigma_h^s(x) + \sigma_h^V(x) \tag{2}$$

As indicated in Section 2.1, the longitudinal strain is monitored in real time by BFOSs. Notably, the total longitudinal strain, including all components (internal pressure, temperature differential and longitudinal bending), is measured. The problem is that the relationship between thermal stress and strain under constrained underground conditions is not simple. In contrast to bending stress, thermal stress is a result of the axial movement of the pipeline being constrained under temperature differential actions; thus, thermal stress in the pipeline is not directly calculated by the measured thermal strain. However, the axial thermal stress is the main contributor to circumferential breaks. Hence, the foremost issue for the digital model is to determine the occurrence and range of the thermal stress in the pipeline.

Under temperature differential actions, axial stress may be created through restrained movement by the friction resistance between the pipe and soil. To prevent circumferential breaks caused by excessive axial thermal stress, devices such as compensators and sliding supports are commonly installed in the pipeline. Then, axial movement occurs at the unfixed end to release axial thermal stress. As a result, the pipeline is divided into two sections, i.e., the anchorage section and transition section (in Fig. 2). The actual anchorage point x_g cannot be determined from the physics-based model because pipe–soil friction has considerable uncertainty. In the proposed sensor networks, the temperatures inside and outside the pipeline can be measured by BFOSs and digital temperature sensors, respectively. By fusing these monitoring data, the temperature differential $\Delta\widetilde{T}$ values (where the tilde $\widetilde{\bullet}$ denotes the actual measurement in this study) are obtained along the pipeline. The temperature differentials provide the signs of the possible axial movements. Once a temperature differential is detected, the difference in the measured strains before and after temperature change is determined as the thermal strain along the pipeline. The section with strain changes is the transition section, while the remaining part without changes can be seen as the anchorage section. According to the above analysis, the axial thermal stress $\sigma_l^T(x)$ can be given by

$$\sigma_l^T(x) = \begin{cases} -E\alpha\Delta\widetilde{T}(x), 0 \le x \le L_g \text{ (anchorage section)} \\ -E\alpha\Delta\widetilde{T}(x) + E\widetilde{\varepsilon}_l^T(x), \mathrm{L}_g \le x \le L \text{ (transition section)} \end{cases} \tag{3}$$

where E and α are the Young's modulus and thermal expansion coefficient of the pipe material, respectively; $\widetilde{\varepsilon}_l^T$ is the thermal strain measured within the transition section; and L_g and L_f are the lengths of the anchorage and transition sections identified by the measured strains, respectively.

If no temperature differentials are detected from the measured data, then the variations in the measured strain are due to external actions such as foundation changes or third-party interference. This strain is identified as longitudinal bending strain $\widetilde{\varepsilon}_l^b$. Then, the longitudinal bending stress is directly calculated using Hooke's law with BFOS data.

$$\sigma_l^b(x) = E \bullet \widetilde{\varepsilon}_l^{\,b}(x) \tag{4}$$

With respect to the strain measurement, according to Kurashima et al. [63], external strain or temperature applied to the fiber leads to the proportionate Brillouin frequency shift (BFS) $\Delta\nu_B$, and temperature compensation is required accordingly [46]:

$$\widetilde{\varepsilon}_l^{\,b}(x) \text{ or } \widetilde{\varepsilon}_l^{\,T}(x) = \frac{\Delta\nu_B - (d\nu_B/dT_B) \bullet \Delta\widetilde{T}_B(x)}{\mathrm{d}\nu_B/d\varepsilon} \tag{5}$$

where x denotes the location along the fiber; $(d\nu_B/d\varepsilon)$ and $(d\nu_B/dT_B)$ denote the strain coefficient and temperature coefficient of the BFOS,

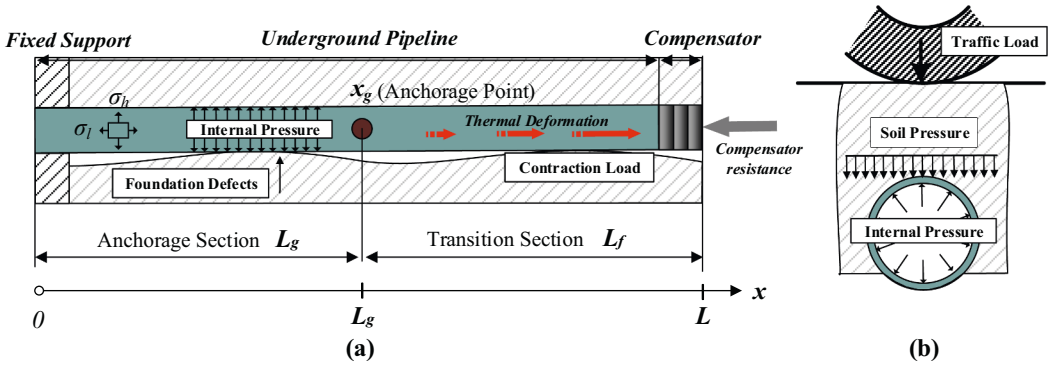


Fig. 2. Underground pipeline stress analysis configuration: (a) longitudinal stress components and (b) hoop stress components.

respectively; and $\Delta\widetilde{T}_B(x)$ is the temperature data acquired by the distributed fiber optic temperature sensor at location x.

When internal pressure is applied to the pipeline, Poisson's effect on the longitudinal stress is determined as follows:

$$\sigma_l^p(x) = \frac{v\widetilde{P}D}{2\delta} \tag{6}$$

where $\widetilde{P}$ is the measured internal pressure; D and δ are the internal diameter and wall thickness of the pipeline, respectively; and v is Poisson's ratio.

By jointly utilizing strain and temperature measurements, the total longitudinal stress σ_l is finally derived by Eqs. (1), (3), (4) and (6).

According to references [34, 52], the components of hoop stress in Eq. (2) are given by the following equations:

$$\sigma_h^p(x) = \frac{\widetilde{P}D}{2\delta} \tag{7}$$

$$\sigma_h^s(x) = \frac{6k_m \bullet C_d\gamma B_d^2 \bullet E\delta r}{E\delta^3 + 24k_d\widetilde{P}r^3} \tag{8}$$

$$\sigma_h^V(x) = \frac{6k_m \bullet I_c C_t F \bullet E\delta r}{L_e\left(E\delta^3 + 24k_d\widetilde{P}r^3\right)} \tag{9}$$

where r is the internal radius of the pipeline; C_d is the coefficient of earth pressure; B_d is the width of the ditch; γ is the unit weight of the soil backfill; k_m is the bending coefficient; k_d is the deflection coefficient; I_c is the impact factor; C_t is a surface load coefficient; L_e is the pipe effective length; and F is the surface wheel load.

By substituting Eqs. (7) to (9) into Eq. (2), the total hoop stress is obtained. Note that all components of hoop stress, as well as σ_l^p, are related to the wall thickness δ. The structural deterioration of the pipeline is mainly due to corrosion. Corrosion induces the loss of pipe wall thickness in a relatively uniform or localized way. Hence, corrosion should be considered in the structural analysis model, especially for lifecycle service. The loss of wall thickness due to local corrosion is commonly obtained by in-line NDT inspection in practical engineering. Once the NDT inspection is performed, the inspection results can be adopted in the proposed framework; accordingly, the stress concentration and localized corrosion growth models should be introduced into the stress calculation at the corrosion pit locations [51]. This study only accounts for the structural deterioration caused by uniform corrosion. Uniform corrosion models have been extensively investigated in many studies on underground and subsea pipelines [64]. A widely used power law model of corrosion [65] was adopted in this study.

$$\delta = \delta_0 - kt^n \tag{10}$$

where δ_0 is the initial wall thickness; t is the exposure period in years; and k and n are the corrosion growth coefficients. This model provides a simplified but practically meaningful expression to capture corrosion growth over time. In the initial phase of service, the parameters k and n can be estimated by field data or experience. With increasing exposure time, the parameters should be validated and updated with inspection data.

By incorporating a corrosion model, a digital model was established for the real-time analysis of pipeline stress. Although it has a solid theoretical basis, the critical loads, i.e., internal pressure and temperature, are both obtained from measured data. Most importantly, the longitudinal bending and axial thermal strains/stresses are directly obtained from measurements. Clearly, this model is different from the classical stress analysis model based on assumed parameters. The proposed approach combines physics-based modeling and data-driven modeling with monitoring data, which enables more informed decision-making and reliable prediction based on a highly interpretable model.

2.2.2. *Reliability-based structural safety assessment*

The proposed digital models involve a physics-based model and monitoring data simultaneously. Several parameters in the models are derived from design values rather than actual values. The structural deterioration process (i.e., corrosion) is associated with a stochastic phenomenon. As a result, uncertainty inevitably exists in material and geometrical properties, the physical models of structural stress analysis, and the monitoring sensors. Aleatory and epistemic uncertainties should be effectively reduced for structural safety assessment and O&M decision-making. In this section, a digital model for pipeline structural assessment and service life prediction based on time-dependent reliability is proposed to quantify the uncertainty during lifecycle service.

The failure condition of pipelines needs to be defined in DTs. Both the Von Mises and Tresca failure criteria are extensively adopted to assess the structural failure of ductile metal pipelines [34], and fracture toughness-based limit states have been proposed to compute the reliability of pipelines with localized defects such as corrosion pits [51]. For urban pipeline networks, strength theory-based approaches (i.e., Von Mises and Tresca) are more appropriate because they seldom perform inner inspections. Without loss of generality, the Tresca failure criterion (i.e., maximum shear stress criterion) is adopted to formulate the time-dependent reliability-based structural safety and prediction model.

$$\sigma_{eq}(x) = \sigma_h(x) - \sigma_l(x) > [\sigma] \tag{11}$$

where σ_{eq} is the equivalent stress and $[\sigma]$ is the allowable stress of the pipe material. The above equation expresses the relationship between the load effects acting on the pipeline and the resistances.

The deterioration process during long-term service may impair the safety and serviceability of pipelines. The probability of failure is defined as the probability of violating the performance function defined by Eq. (11). By assuming that $\sigma_{eq}(x)$ and $[\sigma]$ are statistically independent

random variables with respect to the exposure time t, the (instantaneous) probability of failure at location x is [66].

$$P_F(x,t) = P\{\sigma_{eq}(x,t) - [\sigma] < 0\} = \int_0^{\infty} F_R(u,x,t) f_Q(u,x,t) du \quad (12)$$

where $F_R(u,x,t)$ and $f_Q(u,x,t)$ are the instantaneous cumulative probability distribution function (CPF) of the resistance and the instantaneous probability density function (PDF) of the load effect, respectively. In DTs, the instantaneous probability of failure is adopted to quantify the current stress states along the pipeline by integrating the real-time monitoring data. As a result, the monitoring data can aid in proper assessment, and the probabilistic indicator can handle the randomness induced by the aleatory and epistemic uncertainties. The instantaneous CPF/PDF of failure can be estimated using the first-order reliability method (FORM) or Monte Carlo simulation (MCS).

In addition to real-time assessment, a pipeline DT should have the capacity to predict the service performance over a service or inter-inspection period for proper lifecycle management. A performance indicator is required to address uncertainties associated with time-dependent deterioration development, i.e., corrosion growth [66,67]. A lifetime reliability indicator is introduced in this study to estimate the probability of the time to failure. The time to failure of pipeline T_f is treated as a random variable that defines the elapsed time from installing the pipeline for operation until it fails for the first time [68,69]. The cumulative probability of failure $F(x, T_f)$, the probability that the pipeline at location x is not functioning at time t, is defined as

$$F(x, T_f) = P(x, T_f \le t) = \int_0^t f_T(x,u) du \quad (13)$$

where $f_T(x,u)$ is the PDF of the time to failure derived by performing statistical analysis.

The lifetime reliability measure $F(x, T_f)$ in Eq. (13) cannot be calculated analytically. MCS based on stress analysis and the corrosion growth model is utilized to calculate the PDF of time to failure, and then, the time-dependent probabilistic performance profile is built for the pipeline. The probabilistic performance indicator for service life is estimated, while the remainder of the lifetime is predicted through the statistical information of the monitoring data and pipeline structural analysis model. After obtaining the lifetime reliability profile, a predictive strategy based on the threshold is proposed to schedule the inspection or maintenance time. The decision-making function can be defined according to reliability-based assessment and prediction:

$$T_{inspct} = T_j \quad if\ F(x, T_j) \ge F_{thres}^{inspct} \quad (T_j = 1, 2, \cdots, N_d) \quad (14)$$

where T_{inspct} is the time of the inspection; T_j is the elapsed time (unit: year); N_d is the design service life; and F_{thres}^{inspct} is the threshold of lifetime reliability for the first inspection. Notably, the threshold is determined according to the target reliability indicators for pipeline safety. Based on Eq. (14), the first inspection is planned when the lifetime reliability indicator reaches the predetermined threshold. This strategy is appropriate for newly built structures. With the data collected by the inspection, the parameters of the corrosion growth model can be accordingly validated and updated. The Bayesian approach is commonly adopted for model updating [70,71]. In this study, we only investigate the first inspection plan for newly built pipelines.

According to the time-dependent reliability analysis, digital models for the quantitative assessment and prediction of structural safety are established for pipelines. In the mutual interaction between DTs and monitoring data, the real-time reliability can be obtained to assess the current performance of pipelines. Moreover, lifetime reliability-based prediction enables informed decision-making in pipeline O&M for long-term service. The proposed approach provides effective ways for DT to achieve successful lifecycle management.

2.3. Human–asset interaction interface

DTs enable the real-time sensing, analysis, assessment, and prediction of the structural states of underground pipelines. Undoubtedly, a more efficient human–asset interaction interface is required for DTs. The major obstacle in interactions between the O&M operator and underground pipeline is that the buried asset is not visible, which makes it difficult to effectively find hidden defects and respond quickly to emergencies (i.e., pipe leakages or breaks). The human–asset interaction interface needs to have location-based accessibility and intuitive visualization. AR can allow an operator to have real-time interactions with hidden assets during O&M activities. Extensive investigations have demonstrated that AR is more specifically appropriate for in situ applications (references [72–77]). The implementation of AR as a human–asset interaction interface can add to the long-term value of DTs. In this study, a MAR-based approach is proposed to manage and visualize the data, analysis results and prediction outcomes, which allows a high degree of interaction and immersive visualization with underground pipelines and DTs.

2.3.1. Visualization of geometric and physical attributes

Visualization of geometric appearances is vital for O&M of buried assets. Geometric and physical attributes include the geometry, location, and structural parameters of pipelines. Since pipelines are large-scale, both precision and weight need to be considered for attribute modeling. Highly detailed models can fundamentally ensure precise descriptions, while coarse models can improve modeling efficiency and prevent a single monolithic model from being used for a whole regional pipeline system [78]. Hence, a dynamic hierarchical method based on the Unity3d game engine and Mapbox software development kits (SDKs) is proposed to model the geometric attribute information of underground pipeline systems at multiple spatial scales according to different levels of detail (LODs). Thus, the model is divided into a system-level submodel and a unit-level submodel (in Fig. 3(d) and Fig. 3(e), respectively). The system-level and unit-level models serve different levels of information requirements. For instance, in the process of structural safety assessment, the system-level model describes the association and coupling relationships between assets, and the unit-level model describes the more complex structural state information for each asset.

The system-level geometric attribute model (GAM) consists of pipelines and related facilities (i.e., manholes, pumps, and valves). Regarding the GAM generation of support facilities, the prefabs (Fig. 3 (c)) of a manhole, a pump and a valve are made in Unity3d with reference to their geometric shapes, and then, the geometric models of support facilities are dynamically generated on Mapbox map layers in Unity based on the prefabs and location (L_{at}, L_{on}) collections of cloud-based storage platforms. Finally, their unique identifiers are matched, and basic attributes are added by programming in C#. For pipeline GAM generation at the system level, based on the location $\{(L_{at,\,1}, L_{on,\,1}, D_{ep,\,1}), \cdots, (L_{at,\,s}, L_{on,\,s}, D_{ep,\,s})\}$ and geometry of each pipeline, mesh vertices of the pipeline cross sections are generated at the s nodes by rotating a point n times $(360/n)^\circ$ at a distance r (pipeline radius) in the direction normal to the pipeline segment; then, the mesh model is generated from the mesh vertices, as shown in Fig. 3(d). Furthermore, the unique pipeline identifiers and basic attributes {*material, age, type, operation and*

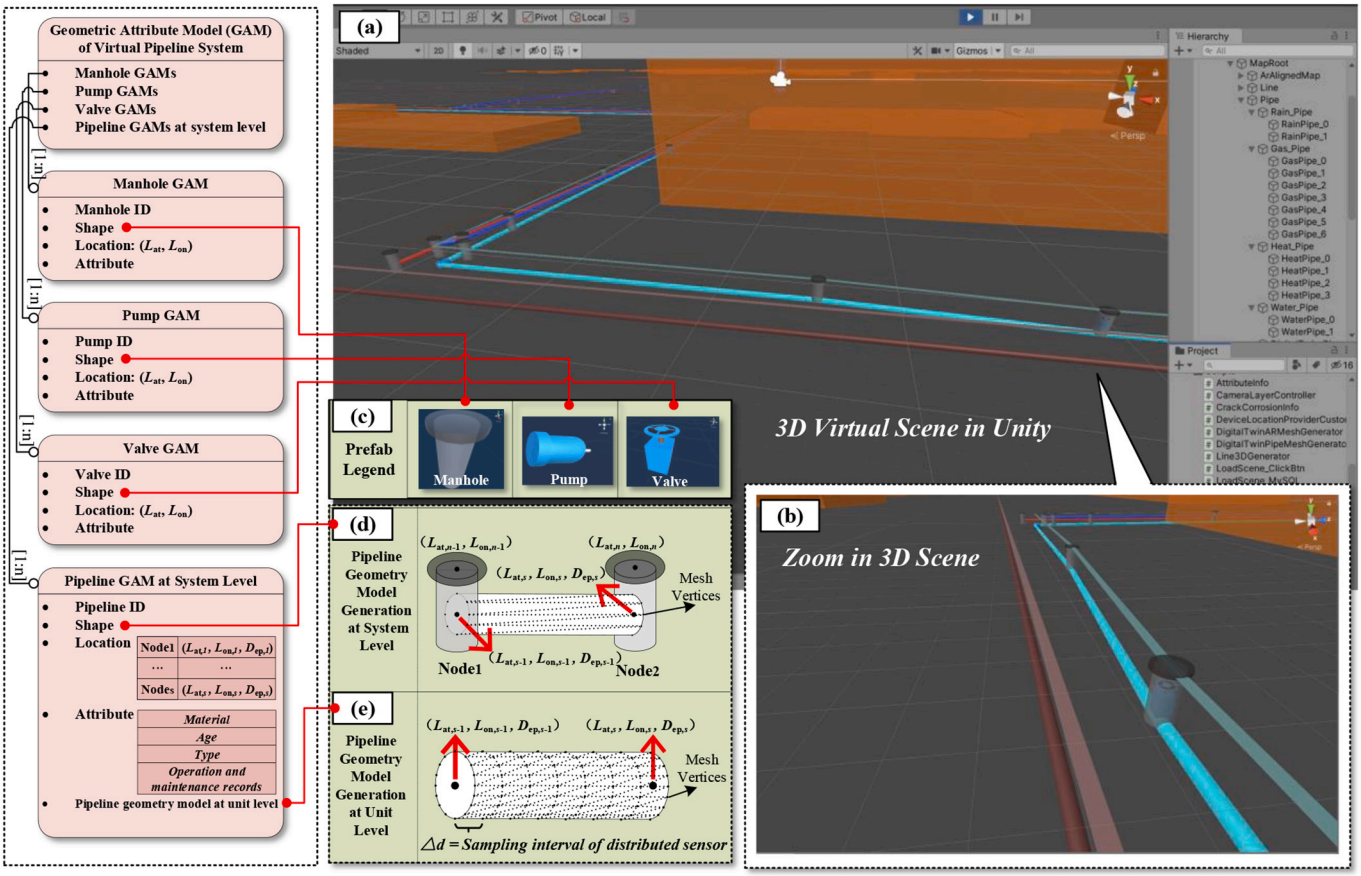


Fig. 3. Geometric attribute model: (a) 3D geometry model dynamic generation in Unity; (b) detailed view of 3D geometry models; (c) prefabs of a manhole, a pump and a valve; (d) pipeline geometry model generation process at the system level; and (e) pipeline geometry model generation process at the unit level.

maintenance records} are added from the infrastructure-as-a-service (IaaS) to rapidly generate lightweight GAMs. The system-level GAMs dynamically constructed in the Unity3d engine are shown in Fig. 3(a) and (b).

For structural monitoring and assessment, the refined pipeline GAM at the unit level should be constructed. The dynamic generation process of the unit-level GAM is similar to that of the system-level GAM, but the pipeline geometry model at the system level is not sufficient to visualize the structural monitoring data, resulting in an imprecise description of the pipeline structural state. Therefore, mesh vertices of the pipeline cross section are generated every Δd (Δd is the sampling interval of the distributed sensors) from the starting node to achieve a high LOD for the model in Fig. 3(e).

2.3.2. Visualization of real-time data and assessment results

The DT should provide operators with real-time structural information (including monitoring raw strain data, stress analysis data, and reliability-based assessment results) to facilitate responding actions or O&M decisions with greater accuracy, efficiency and safety. In addition to visualizing the geometrical appearance, this study presents a 3D nephogram method based on mesh vertices to render the visualization of the monitoring data and assessment results, as shown in Fig. 4.

According to the unit-level GAM, the sampling interval of the distributed sensors is Δd, and thus, the mesh vertices of the pipeline cross section correspond to the monitoring data of the distributed Brillouin optical fiber sensors. The monitoring data are further processed through a joint data–physics driven model (Section 2.2), and the stress or reliability data are visualized through Eq. (15), with the mesh vertices rendered to form a 3D nephogram:

$$\begin{cases} \mathrm{Color}(1,0,0), y(x) \geq y_{1st+}; \\ \mathrm{Color}\left(1, 1-\dfrac{y-y_{2st+}}{y_{2st+}}, 0\right), y_{1st+} > y(x) \geq y_{2st+}; \\ \mathrm{Color}\left(\dfrac{y}{y_{2st+}}, 1, 0\right), y_{2st+} > y(x) \geq y_O; \\ \mathrm{Color}\left(0, 1, \dfrac{y}{y_{2st-}}\right), y_O > y(x) \geq y_{2st-}; \\ \mathrm{Color}\left(0, 1-\dfrac{y-y_{2st-}}{y_{2st-}}, 1\right), y_{2st-} > y(x) \geq y_{1st-}; \\ \mathrm{Color}(0,0,1), y_{1st-} > y(x). \end{cases} \tag{15}$$

where $y(x)$ is the monitoring data and result data $\tilde{\varepsilon}_l^b, \tilde{\varepsilon}_l^T\ (\sigma_l^b, \sigma_l^T, \sigma_{eq})$ or reliability P_F at the distributed monitoring point x; y_{1st+} and y_{1st-} are the first upper and lower thresholds of the datasets, respectively; y_{2st+} and y_{2st-} are the second upper and lower thresholds of the datasets, respectively; and y_O is the zero point of the datasets. When $y(x)$ is the monitoring data and the result data are $\tilde{\varepsilon}_l^b, \tilde{\varepsilon}_l^T\ (\sigma_l^b, \sigma_l^T, \sigma_{eq})$, y_{1st+}, y_{2st+}, y_O, y_{2st-} and y_{1st-} are taken as $max\{y(x)\}_{x=0}^{L}$, $max\{y(x)/2\}_{x=0}^{L}$, 0, $min\{y(x)/2\}_{x=0}^{L}$ and $min\{y(x)\}_{x=0}^{L}$. When $y(x)$ is the reliability P_F, y_{1st+}, y_{2st+}, y_O, y_{2st-} and y_{1st-} are taken as $max\{P_F\}_{x=0}^{L}$, $\overline{\mu} + max\{P_F/2\}_{x=0}^{L}$, $\overline{\mu}$, $\overline{\mu} - min\{P_F\}_{x=0}^{L}$ and $min\{P_F\}_{x=0}^{L}$, respectively, where $\overline{\mu} = mean\{P_F\}_{x=0}^{L}$. $\mathrm{Color}(R,G,B)$ represents the mesh colors of the vertices in the Unity3d

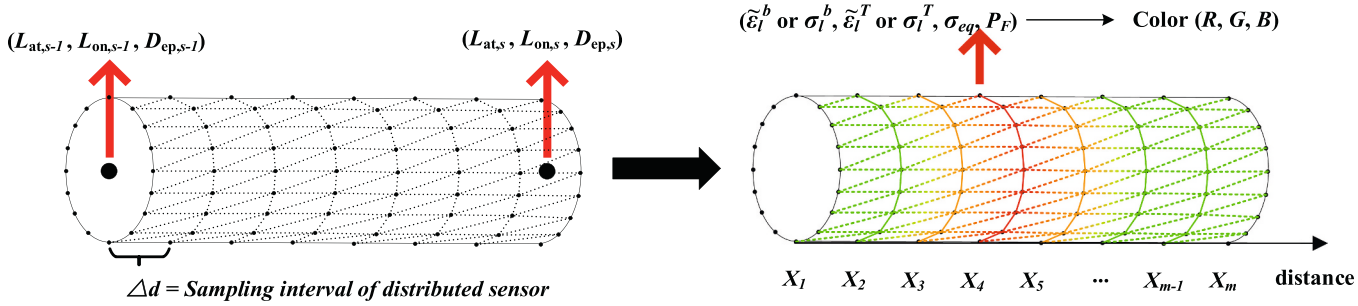


Fig. 4. 3D nephogram form based on the rendering of mesh vertices.

shader in terms of red–green–blue (RGB) values.

2.3.3. Mobile augmented reality visualization

AR is considered to be an effective and fast communication and interaction tool that can visualize virtual objects [79,80]. However, current AR alignment methods, such as quick response (QR) markers [81], simultaneous localization and mapping (SLAM) [82,83] and global positioning systems (GPSs) [28], have two main limitations: (1) the positioning accuracy of currently commercially available SLAM systems and GPSs is far from sufficient for practical engineering requirements, and (2) the application of QR markers is limited to the tabletop level, far from the city level. In addition, the computer vision (CV)-based method [84] is affected by the environment and is mainly used for the appearance of buildings rather than for hidden pipelines. With the rapid development of AR-enabled software (e.g., ARCore, ARKit) and micro-global navigation satellite system (GNSS) surveying receivers with centimeter-level location accuracy (e.g., Lite RTK from Qianxun Spatial Intelligence, Fig. 5(b)), AR-GIS technology can be used to achieve precise mapping of spatial locations between DTs and physical assets in the field (Fig. 5). This solution features lightweight equipment, good precision, environmental robustness and large-scale applicability.

As shown in Fig. 5, GNSSs (e.g., GPS or BeiDou) are used to send signals for navigation and location. Ground-based augmentation systems are responsible for collecting and processing satellite data and providing calibration data to the GNSS receiver in real time. GNSS receivers are used to receive high-precision positioning coordinates using the global positioning system fix data (GGA) format [85]. The mobile app obtains precise positioning coordinates from GNSS receivers in real time via a Bluetooth connection so that the Mapbox map layers are perfectly aligned with the physical world. In addition, ARCore/ARKit's plane anchoring capability (the so-called ARPlaneAnchor object [28]) can be used to "seamlessly" overlay DTs with physical assets. This system continuously searches for feature points to identify the plane of the surroundings and creates ARPlaneAnchor objects to anchor the Mapbox map layers. Alignment of the orientation angle is carried out using a combination of the electronic compass of a mobile device and manual rotation of the map layers [24]. Location-based human–asset interaction is achieved by the above approach.

3. Case study

A case study was performed to investigate the feasibility of the proposed pipeline DT for structural safety assessment. The study area is approximately 83,800 m^2, including gas, heating, water and drainage pipelines as well as inspection manholes, valves and pumps. Fig. 6 and Table 2 show the details of the pipeline system. The hardware and software configurations were first introduced in the DT. Then, MAR-based immersive visualization and in situ interactions were obtained for the pipelines at the system level. The selected pipeline was investigated in detail to demonstrate the capability of the DT for real-time

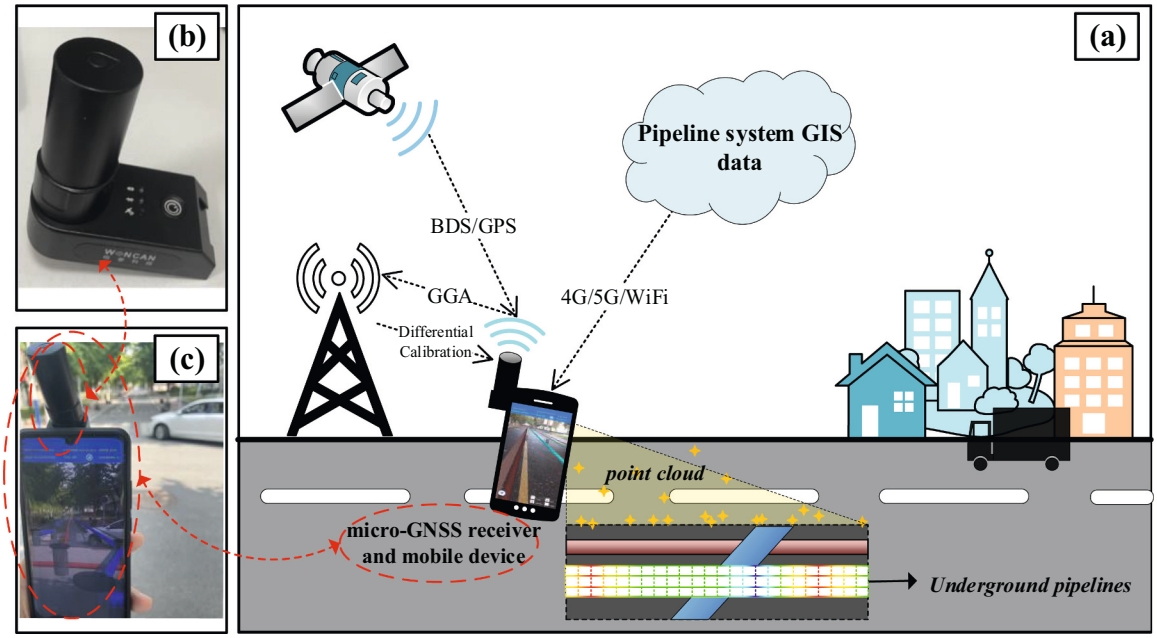


Fig. 5. Mobile augmented reality visualization: (a) the flow process; (b) Lite RTK micro-GNSS surveyor receiver; and (c) portable mobile devices.

safety assessment and lifetime prediction.

3.1. Hardware and software configurations

3.1.1. Sensor networks and instrumentation

The modeling parameters of the pipeline networks were collected from the design documents and as-built surveys. The digital models were then built at the system level. As the pilot demonstration at the unit level for structural safety assessment, GasPipe_6 and HeatingPipe_2 (as shown in Fig. 6 and Table 2) were selected to have distributed sensor networks installed for real-time monitoring of the structural states. These pipeline segments are located below roads with high pedestrian and vehicular traffic. BFOSs were used to monitor the longitudinal strain along the pipeline segments. The commercially available NZS-DSS-C02 and NZS-DTS-C05 were mounted on the pipeline to measure the strain and temperature based on the sensor topology presented in Section 2.1. The sensing fiber core of these sensors is a common SMF-28 single-mode optical fiber with a sensing fiber. Commercially available fiber optic sensors were used in this study. The strain and temperature coefficients were provided by the vender. These coefficients were calibrated by laboratory tests before field application. During sensor installation, the pipe surface was cleaned to remove dirt and grease. Then, high-performance epoxy resin was applied on the top and two sides of the strain sensor to ensure that it was firmly bonded to the surface of the pipe [49]. During this process, the strain sensor was slightly pretensioned to keep it straight along the longitudinal axis of the pipe. Thus, good strain transfer was achieved, and the readings of the strain sensor represented the longitudinal strain of the pipeline. Neubrex NBX-6050A was adopted for real-time data acquisition from the BFOSs. The spatial resolution and sampling interval were set to 10 cm and 5 cm, respectively, and the measurement accuracy was ±10 $\mu\varepsilon$. These BFOSs were arranged along HeatingPipe_2 from the fixed support to the compensator, with a total measurement length of 57.7 m. The measurement length of GasPipe_6 was 61.3 m, and the BFOSs were arranged from the south fixed support to the north fixed support (Fig. 6). Thus, distributed measurements could be obtained along the sensor path. To monitor the internal pressure and temperature, SITRANS P DS III digital pressure transmitters and DS18B20 digital temperature sensors were placed on the pipeline. The SITRANS P DS III digital pressure transmitters were connected to a personal computer (PC) through an analog-to-digital (A/D) converter, while the DS18B20 digital temperature sensors were connected to the PC via an RS-422 data interface. The sampling frequency was 1 Hz. All monitoring data were transmitted to the cloud platform.

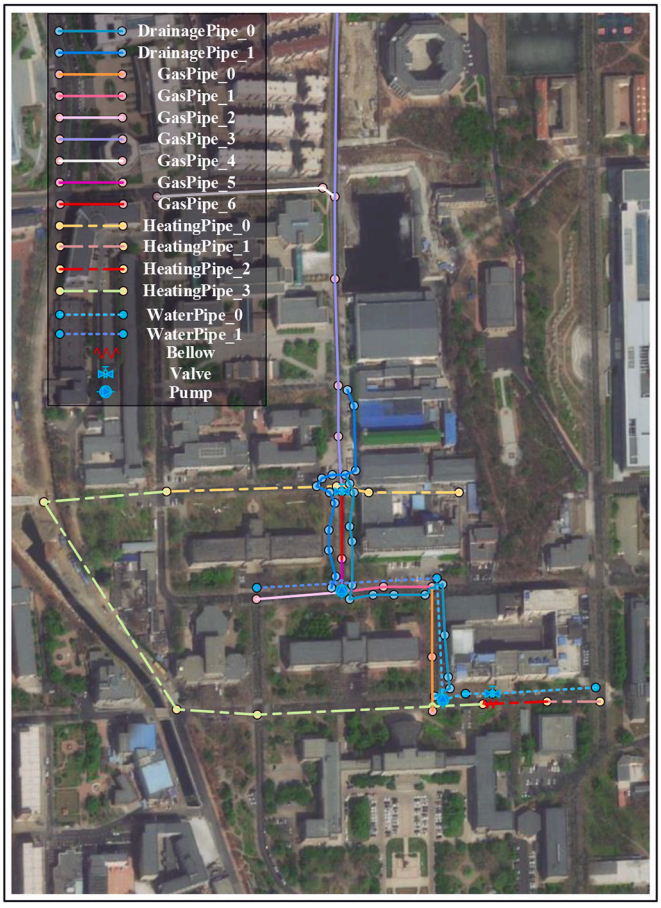


Fig. 6. Underground pipeline system plan of the study area.

3.1.2. DT platform

The Unity3d game engine (version: Unity 2019.4) and C# programming language (version: C#8.0. NET Core 3.0) were used for geometric modeling and joint data-physics driven modeling and were used to launch the final application on a mobile device.

3.1.3. Cloud-based data communication

IaaS is the basic delivery model of cloud computing, and it offers a high level of computing capacity (e.g., networking, servers and data storage) to users [86]. Alibaba Cloud was selected as the IaaS cloud platform for uploading and sending the data collected from the sensor networks. The flexible structured query language (SQL) of the relational database, in combination with 4G/5G/WiFi communication technologies, was adopted to support data create-read-update-delete (CRUD) operations rapidly and securely.

3.1.4. Mobile equipment

A HUAWEI P30 Pro smartphone was selected as the mobile device in this study. The system configuration had the following components: 2 × Cortex-A76 2.6 GHz + 2 × Cortex-A76 1.92 GHz + 4 × Cortex-A55 1.8 GHz central processing units (CPUs), 8 GB random access memory (RAM), and 128 GB read-only memory (ROM). The micro-GNSS receiver was a centimeter-accurate Lite RTK with a 5 Hz update rate, ± (10 + 1 × $10^{-6}D_{bs}$) mm flat accuracy and ± (15 + 1 × $10^{-6}D_{bs}$) mm altitude accuracy (D_{bs} is the distance between the measurement location and the nearby base station). The Lite RTK was mounted on the back of the HUAWEI P30 Pro to accurately track the position of the mobile phone in real time.

3.2. System-level DT service

At the system level, the DT mainly provides real-time information interaction and visualization of underground pipeline systems. In practical applications, a system-level DT superimposes virtual pipeline models on realistic scenarios, which enables the operator to see underground pipelines in the field. The operators can inspect the geometric shapes and geographic locations of the underground pipelines in the immersive perspective view. In this study, areas 1, 2, and 3 in Fig. 7 were used as examples to demonstrate that system-level DTs can improve human awareness of underground pipelines. The system-level geometry models, as shown in Fig. 7 (a), were created in Unity3d, including GIS data. The geometry models as well as the attribute data were then integrated into the MAR-based application. When the operator started the application software, the MAR-based service was launched on site. From the perspective of the user, all the pipelines in the neighboring area were displayed on the screen, as shown in Fig. 7(b). The current location was registered by Lite RTK. The top of the screen displayed the geographical location of the mobile device with longitude and latitude values of 121.52403° and 38.87922°, respectively. The error of horizontal positioning was only 1.4 cm in this test (Fig. 7). The buried depth was shown by selecting a specific pipeline. For instance, by clicking on the #2

Table 2
Details of the physical pipelines.

Pipeline_ID	Type	Location (latitude, longitude, depth)	Attributes	
			Material	Construction Date
DrainagePipe_0	DN400	(38.87923,121.52324,0.5)…(38.88085,121.52231,0.5)	HDPE pipe	08/10/2011
DrainagePipe_1	DN400	(38.88009,121.52231,0.5)…(38.88156,121.52227,0.5)	HDPE pipe	14/10/2011
GasPipe_0	DN300	(38.87921,121.52324,1.0)…(38.88004,121.52321,1.0)	Seamless steel pipe	15/06/2015
GasPipe_1	DN300	(38.88004,121.52321,1.0)…(38.88004,121.52234,1.0)	Seamless steel pipe	17/06/2015
GasPipe_2	DN300	(38.88004,121.52234,1.0), (38.87999,121.52147,1.0)	Seamless steel pipe	20/06/2015
GasPipe_3	DN300	(38.88084,121.52231,1.0)…(38.88491,121.52233,1.0)	Seamless steel pipe	27/06/2015
GasPipe_4	DN300	(38.88305,121.52221,1.0)…(38.88304,121.52080,1.0)	Seamless steel pipe	28/06/2015
GasPipe_5	DN300	(38.88004,121.52234,1.0), (38.88032,121.52233,1.0)	Seamless steel pipe	23/06/2015
GasPipe_6	DN300	(38.88032,121.52233,1.0), (38.88084,121.52231,1.0)	Seamless steel pipe	24/06/2015
HeatingPipe_0	DN350	(38.88084,121.52324,1.6)…(38.88084,121.52147,1.6)	Seamless steel pipe	20/08/2017
HeatingPipe_1	DN350	(38.87921,121.52482,1.6), (38.87921,121.52428,1.6)	Seamless steel pipe	04/08/2017
HeatingPipe_2	DN350	(38.87921,121.52428,1.6), (38.87920,121.52362,1.6)	Seamless steel pipe	09/08/2017
HeatingPipe_3	DN350	(38.87920,121.52362,1.6)…(38.88084,121.52147,1.6)	Seamless steel pipe	18/08/2017
WaterPipe_0	DN350	(38.87924,121.52482,1.2)…(38.87922,121.52326,1.2)	Ductile iron pipe	08/10/2011
WaterPipe_1	DN350	(38.87922,121.52326,1.2), (38.88005,121.52322,1.2)	Ductile iron pipe	15/10/2011

heating pipeline, the selected pipeline was highlighted with a flashing edge (left diagrams of Fig. 7(c)). The pipe ID and depth were then shown as HeatPipe_2 and − 1.6 m, respectively. By moving the device, the positions were tracked, and the pipelines were accordingly displayed in the field of view of the screen. From Fig. 7(b), the pipelines and their related facilities were correctly localized and immersively visualized in each area. The relative locations among the pipelines could be pinpointed from the perspective of the operator. In underground construction or rehabilitation actions, the existing pipelines can be accurately mapped in the field by using this MAR-based location service, which is a promising tool for preventing underground pipeline strikes.

In addition to in situ mapping of underground pipelines, the system-level DT enables information querying in a friendly, interactive way. The attribute data can be queried by clicking the pipeline. The nominal diameter, material, type, and age are displayed on the screen. The O&M records, including content, time and operator, are shown to support rational operation. The detailed information provided by the DTs is magnified in the bottom right diagram of Fig. 7(c). The MAR-based interaction enhances human perception and information extraction from the virtual world to the real world, thus improving access to and utilization of information. By utilizing multisource heterogeneous data, pipeline DTs can guide various real-life project tasks, such as communication among different teams, project management and unbiased decision-making.

3.3. Unit-level DT service

As mentioned in Section 3.1, the pipeline with ID HeatingPipe_2 was selected as the testbed to investigate the unit-level DT. The geometry, material and load parameters of the pipeline are listed in Table 3. The pipeline segment is subjected to altered loads and has a thermal compensator. In comparison with the water and gas pipelines, this pipeline suffers from large temperature differentials. The pipe–soil interaction is more complicated during long-term service. The unit-level DT of the pipeline segment was established by integrating the real-time monitoring data with the joint data–physics driven models proposed in Section 2.2. In the MAR-based application, the unit-level DT is entered by clicking the *DigitalTwin* button (see Fig. 7(c)). At this level, the monitoring data and stress analysis results are visualized for in situ data query, which makes the preliminary assessment of the pipeline segment intuitive. Then, the reliability-based structural safety assessment and lifetime prediction are obtained from the DT to support O&M decision-making. All the assessments are performed in the MAR environment in real time.

3.3.1. Data visualization and real-time assessment

HeatingPipe_2 was installed on August 9, 2017. The baseline data of the longitudinal strain were collected with BFOSs. After trench backfill and road pavement application (from August 21, 2017, to September 28, 2017), bending deformation occurred because of the differential settlement along the pipeline segment. Then, the bending strains were monitored by comparing the current data with the baseline data. When the operator enters the "Query Date" as "2017_09_28" and selects the data type, the raw data (e.g., longitudinal strain or temperature) or the derived data (e.g., bending strain or equivalent stress) are displayed in the MAR-based DT. The visualization is presented in two ways: a 3D nephogram and a 2D curve. In Fig. 8 (a) and (c), the bending strains are displayed in nephogram form. The different colors represent the different amplitudes of bending strain. The distribution of bending strain along the pipeline segment is displayed in an intuitive way. According to the bending strain nephogram, the high strain sector in the pipeline can be easily localized with the aid of MAR. Furthermore, the curves of bending strain and temperature as well as a magnified diagram are also shown in the DT. The temperatures were uniformly distributed with an amplitude of 23 °C. This indicates that no leakage occurred along the pipeline. However, the bending behavior was very complicated, with tensile and compressive bending alternatively occurring in the pipeline. Two peaks appeared at locations of 2.4 m and 56.3 m with amplitudes of 625 $\mu\varepsilon$ and 938 $\mu\varepsilon$, respectively. Based on the O&M records of the system-level DT service in Fig. 7 (c), during the construction phase, the pipeline is subject to bending deformation due to the overburden loads, unevenness of the pipe trench and foundation settlement [49]. Such a large bending strain and its complicated distribution cannot be predicted in advance. The monitoring data demonstrate that the proposed sensing scheme is able to provide the bending strain for the pipeline DT in real time. Moreover, MAR-based data visualization can add the values of monitoring data for use and interpretation.

The pipeline experienced a heating and boosting process from October 28, 2017, to November 4, 2017. The internal temperature increased from 23 °C to 57 °C, as measured by the DS18B20 digital temperature sensors, while the internal pressure rose to 1.0 MPa, as monitored by a pressure transmitter. At the same time, longitudinal strain was monitored by the BFOSs. The longitudinal strain caused by the process is shown by the nephogram and curves in the DT (in Fig. 8 (b) and (d)). The measured strain did not present a clear bending feature, but instead a two-section shape. By fusing the internal temperature and pressure data, this was recognized as axial thermal strain because a thermal compensator (i.e., bellows) was set in the right end of the pipeline segment. The position of the bellows could be identified by using the system-level DT. For comparison, the longitudinal strain was measured 72 h later. As shown in Fig. 8(d), the longitudinal strain after 72 h was almost a flat line (i.e., dark blue line) with very small fluctuations. No bending or axial deformation occurred during the normal operation phase. The amplitudes of the dark red curve (T1) before 21.77

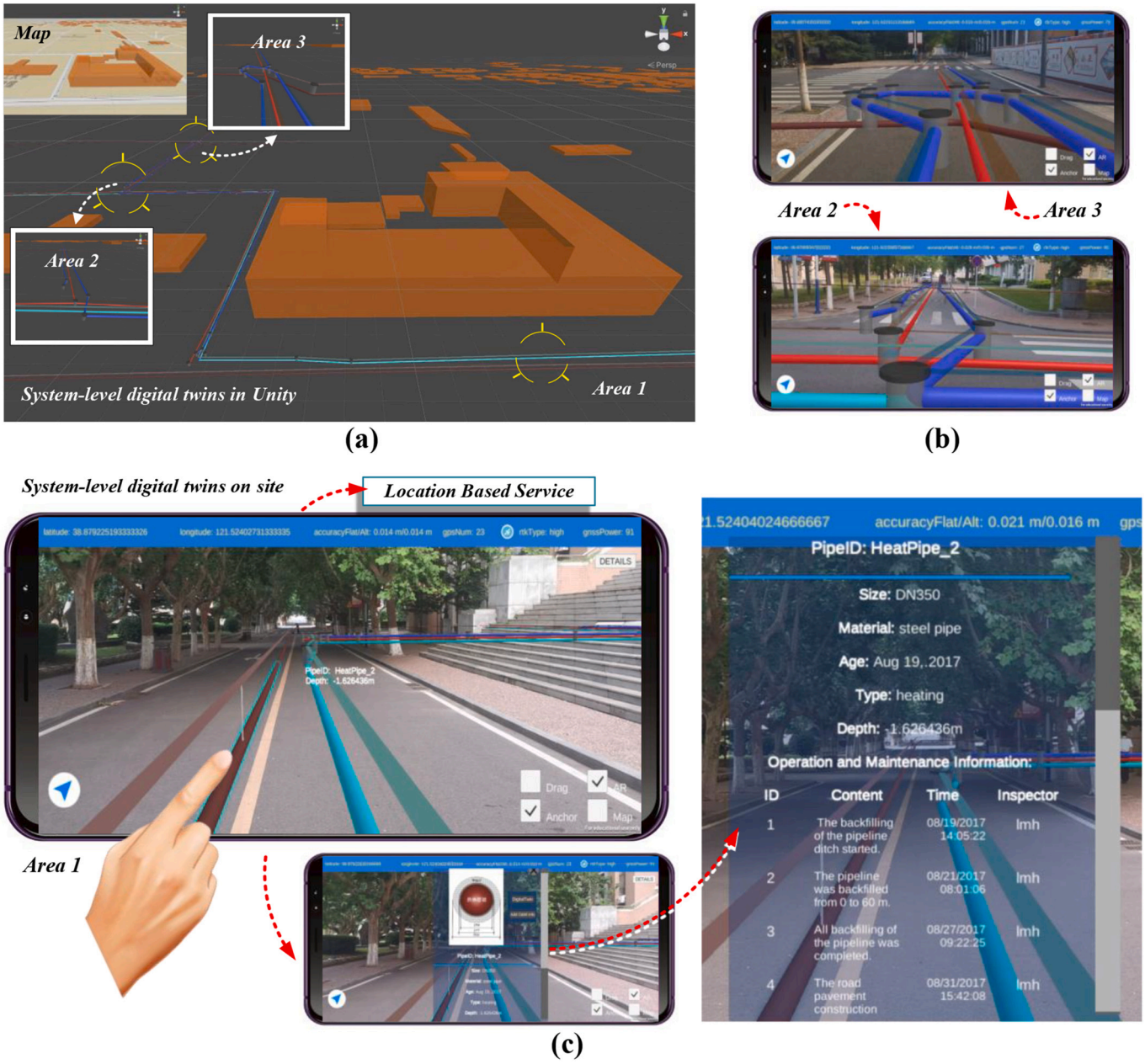


Fig. 7. Location-based service for system-level DTs: (a) aerial view of the pipeline system in Unity; (b) in situ view for the inspector; and (c) comprehensive information query for underground pipelines in the field.

m were approximately 0. Thus, this part could be identified as the anchorage section, while the measured strains from 21.77 m to 57.70 m demonstrated a gradually increasing trend that could be approximately fitted by a straight line, and this part was seen as the transition section. Notably, the complex soil–pipe interaction (induced by uneven bedding, imperfect friction, etc.) was nonuniform along the pipeline, leading to T1 not being an ideal linear curve in the transition section. To the best of our knowledge, field observations of axial thermal deformation in underground pipelines have rarely been reported in the literature. It is evident that thermal deformation cannot be obtained without BFOSs.

When the bending strain and axial thermal strain were obtained, all the monitoring data (including the internal pressure and temperature differential) and the design parameters were input into the joint data–physics driven stress analysis models. The longitudinal and hoop stresses at arbitrary locations in the monitoring segment were calculated. To assess structural safety, the longitudinal bending stress is the primary stress required to balance the external loads, resulting in an allowable stress that is never permitted to be higher than σ_y on the outside wall of the pipe [87]. As shown in Fig. 8 (e), the peak longitudinal bending stress of HeatingPipe_2 was 193 MPa, which did not reach the yield strength (235 MPa) of the pipe material. According to the ASME Code [87] and Chinese Code [88], thermal stress is classified as the secondary stress developed by the self-constraint of the adjacent material or the structure, and it should be analyzed using shakedown theory. Therefore, a range of stress variation larger than the yield strength can occur in the case of self-limiting thermal deformation [89]. According to the design code [88], the allowable value $[\sigma]$ in Eq. (11) should be $min\{2\sigma_y, \sigma_b\} = 375$ MPa. As shown in Fig. 8 (f), the maximum equivalent stress was 262 MPa at 2.36 m in the anchorage section, while the maximum value was 295 MPa at 56.77 m in the transition section. The peak equivalent stresses were all less than the allowable value. The pipeline segment can be preliminarily determined to be in a safe state.

Table 3
Random variables and corresponding parameters.

Type of variables	Symbol	Description	Mean value	Coefficient of variation Real time/ Lifetime
Material	E	Modulus of elasticity	206 (GPa)	0.033
	σ_y	Yield strength	235 (MPa)	0.05
	σ_b	Tensile strength	375 (MPa)	0.05
Geometry	D	Nominal diameter	350 (mm)	0.01
	δ_0	Wall thickness	9 (mm)	0.05
	L_e	Pipe effective length	–	0.03
	B_d	Width of ditch	–	0.05
Coefficients	C_d	Calculation coefficient	1.32	0.2
	I_c	Impact factor	1.25	0.2
	C_t	Surface load coefficient	0.12	0.15
	α	Thermal expansion coefficient	12.6×10^{-6} ($^{\circ}C^{-1}$)	0.1
	ν	Poisson's ratio	0.3	0.023
	k	Multiplier constant	0.226	0.56
	n	Exponential constant	0.53	0.26
	k_m	Bending moment coefficient	0.235	0.15
	k_d	Deflection coefficient	0.108	0.15
	γ	Unit weight of soil	1.89×10^{-5} (N/mm^3)	0.1
Loading	F	Wheel load of traffic	41.2×10^3 (N)	0.1
Monitoring	$\widetilde{P}$	Internal pressure	–	0.05/0.1
	$\Delta\widetilde{T}$	Temperature differential	–	0.05/0.15
	$\widetilde{\varepsilon}_L$	Longitudinal bending strain	–	0.01/0.01

Note: The random variables all obey a normal distribution, and the statistics are given by [34,51,53].

However, Fig. 8(f) shows that the equivalent stress derived from the joint data–physics driven model was greater than that from the common calculation model. The commonly used method underestimates the stress because the actual bending strain cannot be obtained. The results based on in situ monitoring indicate that the actual stress state of the pipeline might be dominated by the bending actions due to differential settlements. Therefore, the BFOS-based DT can achieve a more accurate estimation of the real stress state for the underground pipeline.

The uncertainties related to the models, parameters, and sensors were considered in the reliability-based structural safety assessment. For the DT, the reliability computation was performed by using the FOSM-based JC method (i.e., the improved first-order second-moment method), and the analysis results are shown in Fig. 9(a) with a reliability index nephogram. The MAR-based interaction interface showed the real-time reliability index of the pipeline. Most of the reliability indices along the pipeline were >6.0. The reliability indices at the peak stress locations were reduced to 4.74 and 3.51; however, these values were greater than the target reliability index of 3.2 specified by the design code [90]. The real-time reliability showed that the pipeline was globally safe after the heating and boosting process. The reliability-based method provides structural safety assessment in a probabilistic sense by accounting for the inevitable uncertainties.

A comparison of the JC method and MCS method for computing the real-time reliability is shown in Fig. 9(b). This demonstrates that compared with the MCS method, the JC method demonstrates acceptable accuracy of the calculation results, with an average error of 0.18%, but saves significant time costs (MCS: 5.9 h; JC: 1.7 s) and makes real-time assessment possible. Precise online assessment can be achieved in pipeline DTs for early warning and timely control of emergencies.

3.3.2. *Service life prediction and detection/maintenance strategy*

For long-term service, predictive O&M must be performed with the aid of a pipeline DT. With increasing service time, corrosion plays an increasingly important role in structural deterioration. Based on real-time monitoring and a corrosion growth model, the lifetime reliability of the pipeline was analyzed. As previously mentioned, the most dangerous locations in the anchorage section and transition section were at 2.36 m and 56.77 m, respectively, as obtained from the real-time assessment. Since these locations are more susceptible to damage that would lead to pipeline failure, they were selected for lifetime reliability-based failure analysis. By performing an MCS with 250,000 samples, the evolutionary process of the PDF of the time to arrive at the limit state of allowable stress is plotted in Fig. 10. The PDF of the time, as well as the mean and standard deviation of time t, needed to reach various equivalent stresses can be obtained from Fig. 10(a). The evolution processes with and without bending stress are totally different. In Fig. 10(b), the PDFs of the time to failure at the most dangerous locations in the anchorage and transition sections are obtained, and they have a higher probability density in the interval [0, 50] than that without accounting for the bending stress. The probability of failure for a specific service duration (e.g., 50 years) can be estimated by integrating the PDF from 0 to this time period.

Based on the lifetime reliability analysis, the DT presents a lifetime performance profile of the pipeline throughout its service life, as shown in Fig. 11. Then, the predictive O&M strategy can be recommended. In this study, only the first inspection schedule is investigated. The basis of the inspection plan is threshold-based. The threshold that triggered the first inspection was assumed to be 10^{-2} (similar to that in [51,91]). When the CPF reached 10^{-2}, the corresponding time of the first inspection was 9 years. If the bending stress was neglected, then the counterpart increased to 17.2 years. There was an 8.2-year difference. The DT provides a more prudent decision basis from the monitoring data. Before the first inspection, once the changes are detected by the distributed sensor networks, the structural state should be reassessed by using the new data, and the lifetime reliability should be updated accordingly. When the first inspection is completed, the inspection data of wall thinning are used to validate the corrosion growth model. Subsequently, the Bayesian method can be adopted to update the parameters (i.e., *k and n*) in the corrosion growth model. From the perspective of the lifecycle, the DT can dynamically assess and predict the structural state of the pipeline in a self-evolving manner.

4. Conclusions

DTs pertain to sensing, assessment, prediction and control of structural states through the real-time mutual interaction between the physical asset and its DT. This paper presents BFOS- and MAR-based methodologies to develop DTs for underground pipelines. Sensor networks integrating BFOSs and IoT-based digital sensors were proposed to enable real-time monitoring of the structural states of underground pipelines. In the framework of digital modeling, a joint data–physics

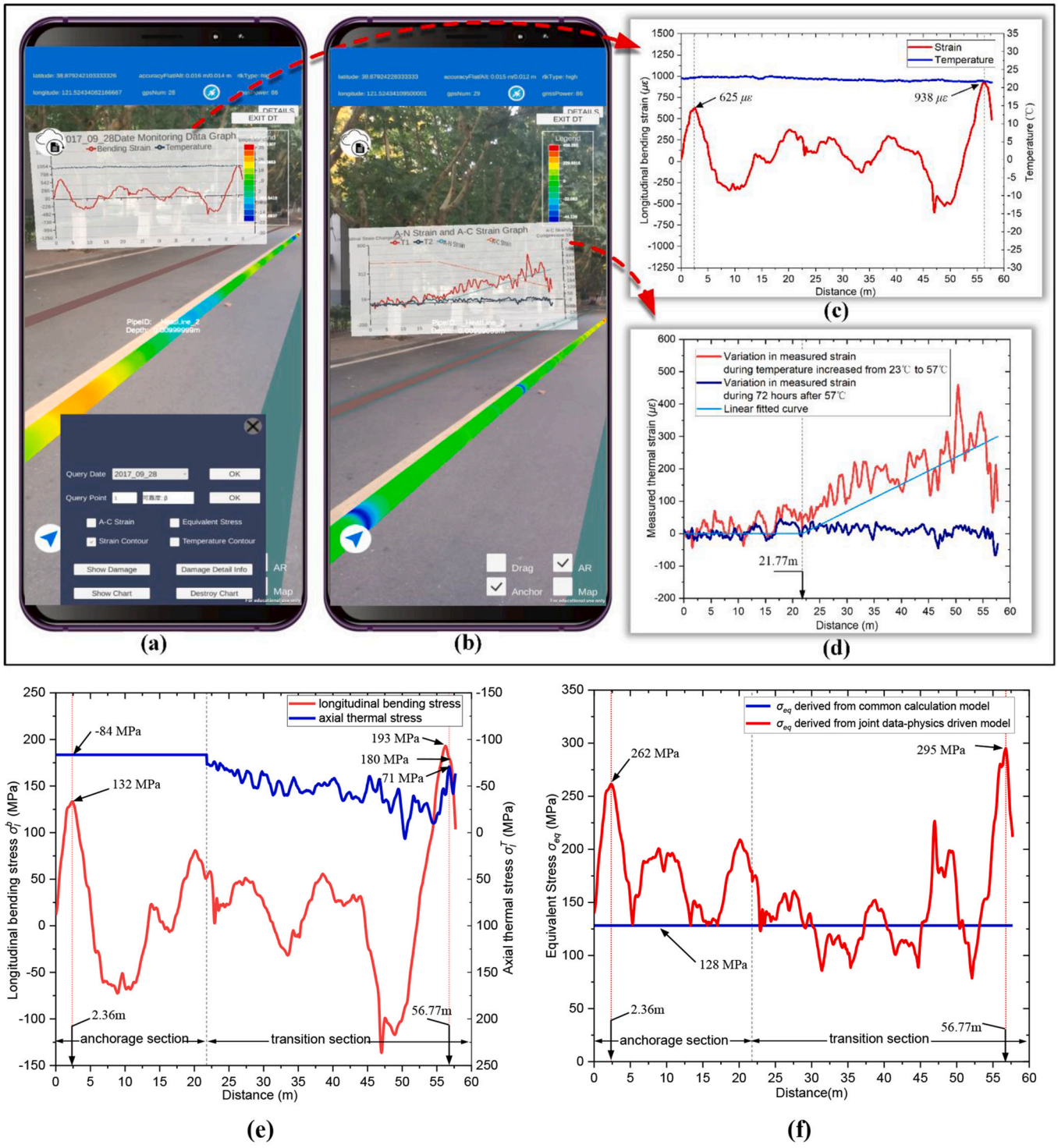


Fig. 8. AR curves and 3D nephogram of the monitoring data and result data: (a) 3D nephogram of bending strain $\tilde{\varepsilon}_l^b$; (b) 3D nephogram of measured thermal strain $\tilde{\varepsilon}_l^T$; (c) curves of bending strain $\tilde{\varepsilon}_l^b$; (d) curves of measured thermal strain $\tilde{\varepsilon}_l^T$; (e) curves of longitudinal bending stress σ_l^b and axial thermal stress σ_l^T; and (f) curves of equivalent stress σ_{eq}.

driven model was proposed to assess the structural stress state in a distributed manner, and reliability-based structural safety assessment models were built for lifetime prediction and informed O&M decisions. A MAR-based human–asset interaction interface was developed to provide real-time interactions between the operator and the hidden pipelines. A series of field experiments were carried out to investigate the feasibility of the proposed methodology.

Based on the case study, the following conclusions can be drawn:

- The proposed sensor networks play an important role for the DT. The unpredictable deformations, e.g., longitudinal bending strain and axial thermal strain, along the pipeline were measured. The results of the case study showed that the bending response caused by overburden loads, the unevenness of the pipe trench and foundation

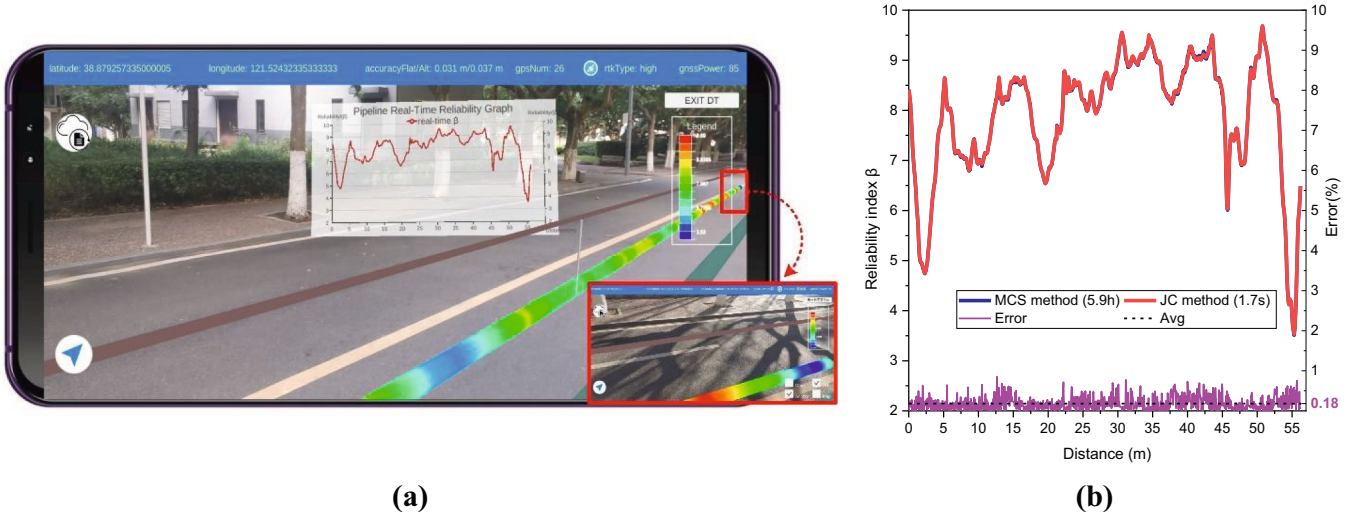


Fig. 9. Real-time safety assessment: (a) AR graph and 3D nephogram and (b) comparison between the JC method and MCS method (sample size is 250,000).

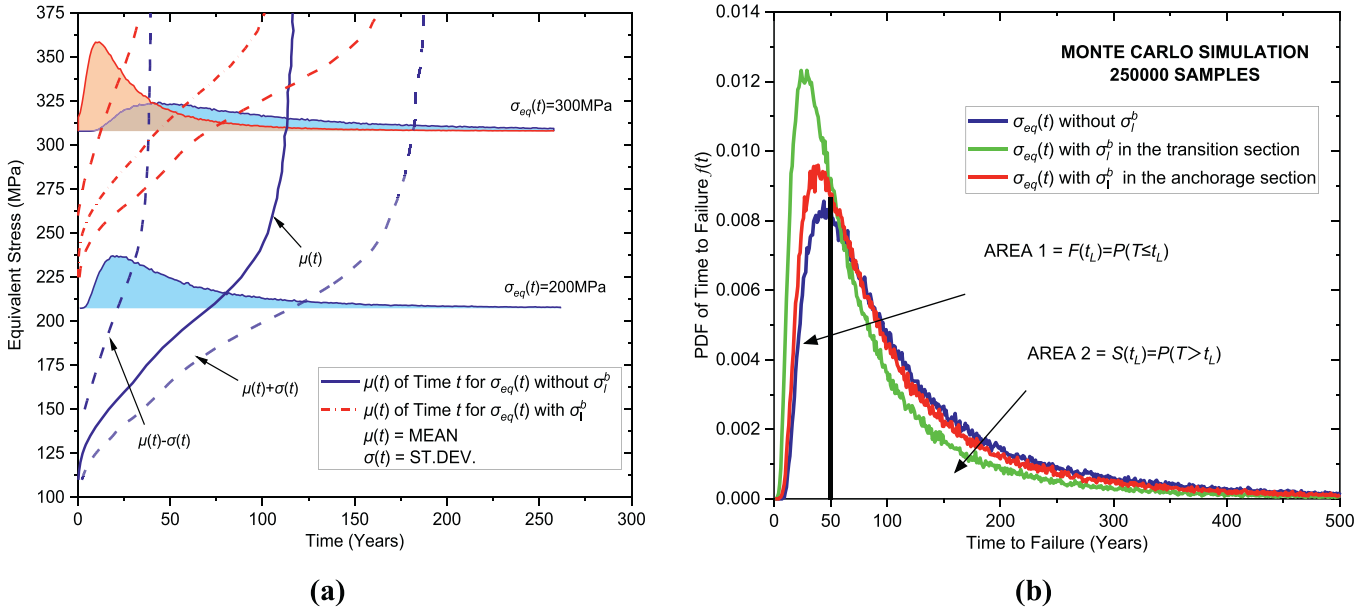


Fig. 10. Lifetime measures of the #2 heating pipeline: (a) time-dependent equivalent stress with PDFs of time when $\sigma_{eq}(t)$= 200 MPa and 300 MPa and (b) PDF of the time to failure.

settlement reached 80% of the allowable value (i.e., yield strength), which could be a main failure factor for the underground pipeline. The measurements of larger-than-expected bending stress induced by several conditions during the construction and early service phases offer field evidence for updating the common assumptions of current calculation models.

- The joint data–physics driven model can accurately simulate the actual stress state in the pipeline by integrating the monitoring data and the well-understood structural analysis theory. The case results showed that the equivalent stress derived from the joint data–physics driven model, benefiting from the integration of monitoring data, was greater than that from the common calculation model. Reliability-based assessment and prediction can address the uncertainties associated with the models, parameters and sensors to produce more rational results and optimize the inspection schedule. The case study found that the corresponding time of the first inspection predicted by the DT was 9 years, which was 8.2 years earlier than that of the common calculation model that cannot account for the actual structural stress state.
- A MAR-based approach was proposed to manage and visualize the data, analysis results and prediction outcomes of pipelines. In the human–asset interface, both virtual and real worlds coexist, and immersive visualization allows hidden underground pipelines to be seen in the field. The case results found that operators experienced a new and improved human–asset interaction in which pipeline location distributions and dangerous locations of structural states were visualized to assist with construction and maintenance. The proposed human–asset interaction interface can achieve easier, more intuitive and more efficient O&M for underground pipelines.

The proposed DT provides a promising tool for the lifecycle management of underground pipelines. However, the present study requires the installation of BFOSs on the pipeline, and thus, it is limited to newly built pipelines. The proposed stress calculation models are applicable to

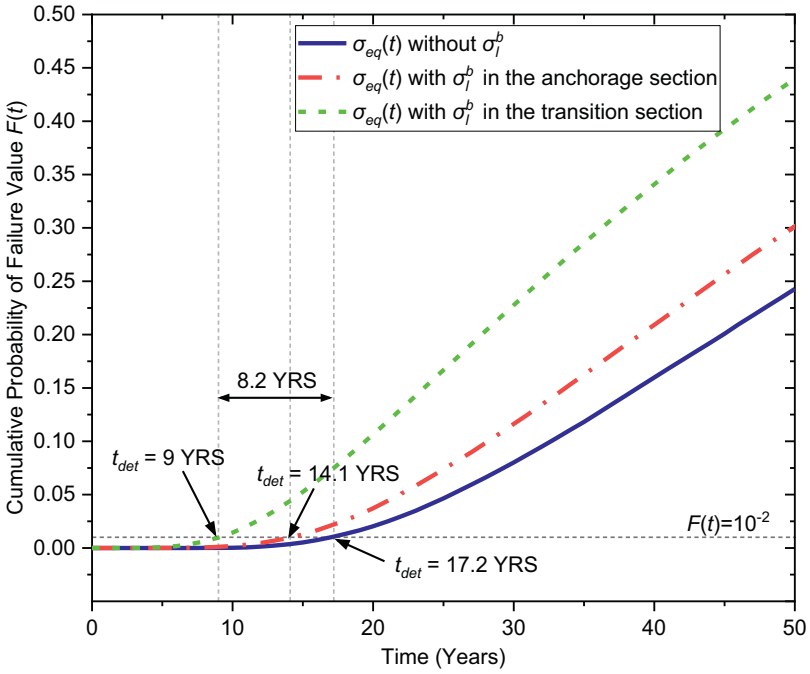


Fig. 11. Cumulative probability of failure and probabilistic maintenance planning.

continuous thin-walled metallic pipelines with diameters <500 mm, and the scope of this study does not include bell-and-spigot pipes. Moreover, the current study only concerns the decision basis for the first inspection according to the uniform corrosion model. In the future, in-line robotic systems and localized damage models should be further adopted for lifecycle O&M by accounting for more complicated deterioration scenarios (i.e., corrosion pits, cracks, etc.).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

[1] D. Liu, J. Chen, S. Li, W. Cui, An integrated visualization framework to support whole-process management of water pipeline safety, Autom. Constr. 89 (2018) 24–37, https://doi.org/10.1016/j.autcon.2018.01.010.
[2] T. Zheng, Z. Liang, L. Zhang, S. Tang, Z. Cui, Safety assessment of buried natural gas pipelines with corrosion defects under the ground settlement, Eng. Fail. Anal. 129 (2021) 1–14, https://doi.org/10.1016/j.engfailanal.2021.105663.
[3] M. Grieves, J. Vickers, Digital twin: Mitigating unpredictable, undesirable emergent behavior in complex systems, in: F.-J. Kahlen, S. Flumerfelt, A. Alves (Eds.), Transdisciplinary Perspectives on Complex Systems: New Findings and Approaches, Springer International Publishing, Cham, 2017, pp. 85–113, https://doi.org/10.1007/978-3-319-38756-7_4.
[4] T.H.J. Uhlemann, C. Lehmann, R. Steinhilper, The digital twin: realizing the cyber-physical production system for industry 4.0, Procedia CIRP 61 (2017) 335–340, https://doi.org/10.1016/j.procir.2016.11.152.
[5] M. Liu, S. Fang, H. Dong, C. Xu, Review of digital twin about concepts, technologies, and industrial applications, J. Manuf. Syst. 58 (2021) 346–361, https://doi.org/10.1016/j.jmsy.2020.06.017.
[6] F. Tao, Q. Qi, Make more digital twins, Nature 573 (7775) (2019) 490–491, https://doi.org/10.1038/d41586-019-02849-1.
[7] Gartner, Prepare for the Impact of Digital Twins, from. https://www.gartner.com/smarterwithgartner/prepare-for-the-impact-of-digital-twins, 2017. Accessed date: 25 April 2019.
[8] M. Shafto, M. Conroy, R. Doyle, E. Glaessgen, C. Kemp, J. LeMoigne, L. Wang, Modeling, Simulation, Information Technology & Processing Roadmap, Technology Area 11, Washington, in: https://sbir.nasa.gov/content/modeling-simulation-information-technology-and-processing, 2012.
[9] C. Ye, L. Butler, B. Calka, M. Iangurazov, Q. Lu, A. Gregory, M. Girolami, C. Middleton, A Digital Twin of Bridges for Structural Health Monitoring, Proceedings of the 12th International Workshop on Structural Health Monitoring, Stanford University, 2019, pp. 3–10, https://doi.org/10.12783/shm2019/32287.
[10] Q. Lu, A.K. Parlikad, P. Woodall, G. Don Ranasinghe, X. Xie, Z. Liang, E. Konstantinou, J. Heaton, J. Schooling, Developing a digital twin at building and city levels: case study of West Cambridge campus, J. Manag. Eng. 36 (3) (2020) 1–19, https://doi.org/10.1061/(asce)me.1943-5479.0000763.
[11] I. Errandonea, S. Beltrán, S. Arrizabalaga, Digital twin for maintenance: a literature review, Comput. Ind. 123 (2020) 1–14, https://doi.org/10.1016/j.compind.2020.103316.
[12] T.R. Wanasinghe, L. Wroblewski, B.K. Petersen, R.G. Gosine, L.A. James, O. De Silva, G.K.I. Mann, P.J. Warrian, Digital twin for the oil and gas industry: overview, research trends, opportunities, and challenges, IEEE Access 8 (2020) 104175–104197, https://doi.org/10.1109/access.2020.2998723.
[13] C. Boje, A. Guerriero, S. Kubicki, Y. Rezgui, Towards a semantic construction digital twin: directions for future research, Autom. Constr. 114 (2020) 1–16, https://doi.org/10.1016/j.autcon.2020.103179.
[14] K. Lin, Y.-L. Xu, X. Lu, Z. Guan, J. Li, Digital twin-based collapse fragility assessment of a long-span cable-stayed bridge under strong earthquakes, Autom. Constr. 123 (2021) 1–13, https://doi.org/10.1016/j.autcon.2020.103547.
[15] N. Mohammadi, J.E. Taylor, Smart City Digital Twins, 2017 IEEE Symposium Series on Computational Intelligence (SSCI), 2017, pp. 1–5, https://doi.org/10.1109/SSCI.2017.8285439.
[16] Q. Lu, X. Xie, A.K. Parlikad, J.M. Schooling, Digital twin-enabled anomaly detection for built asset monitoring in operation and maintenance, Autom. Constr. 118 (2020) 1–16, https://doi.org/10.1016/j.autcon.2020.103277.
[17] E. Fowler, T. Salazar, Think smart, pump smart, World Pipelines 7 (2018) 63–66. https://www.worldpipelines.com/magazine/world-pipelines/july-2018.
[18] M. Xiong, L. Gu, Z. Wu, Y. Deng, S. Li, Y. Zou, Z. Li, C. Feng, Construction and application of digital twin in the in-service oil and gas pipeline, Oil and Gas Stor. Transportat. 38 (5) (2019) 503–509. https://kns.cnki.net/kcms/detail/13.1093.te.20190214.0851.002.html.
[19] V.A. Stennikov, A. Alekseev, E. Mikhailovsky, N.I. Voropai, S.V. Alekseenko, S. P. Filippov, N.A. Yusifbeyli, B. Sereter, P. Changwei, F.J. Lin, M. Negnevitsky, C. Rehtanz, J.Y. Yoon, Development of information and computing technologies for modelling of pipeline and hydraulic systems, in: E3S Web of Conference: ENERGY-21- Sustainable Development & Smart Management, Sept. 7, 2020-Sept.

11, 2020, EDP Sciences, Irkutsk, Russia, 2020, pp. 1–9, https://doi.org/10.1051/e3sconf/202020902002.
[20] J. Lieberman, A. Leidner, G. Percivall, C. Ronsdorf, Using big data analytics and IoT principles to keep an eye on underground infrastructure, in: 2017 IEEE International Conference on Big Data (Big Data), Dec. 11, 2017-Dec. 14, 2017, IEEE, Boston, MA, U.S, 2017, pp. 4592–4601, https://doi.org/10.1109/BigData.2017.8258503.
[21] W.O. Carvalho, BIM and AM to manage critical and relevant water and wastewater utilities assets, in: E. Toledo Santos, S. Scheer (Eds.), Proceedings of the 18th International Conference on Computing in Civil and Building Engineering, Aug. 18, 2020-Aug. 20, 2020, Springer International Publishing, Cham, 2020, pp. 697–720, https://doi.org/10.1007/978-3-030-51295-8_49.
[22] G. Schall, S. Zollmann, G. Reitmayr, Smart Vidente: advances in mobile augmented reality for interactive visualization of underground infrastructure, Pers. Ubiquit. Comput. 17 (7) (2012) 1533–1549, https://doi.org/10.1007/s00779-012-0599-x.
[23] X. Zhang, Y. Han, D. Hao, Z. Lv, ARGIS-based outdoor underground pipeline information system, J. Vis. Commun. Image Represent. 40 (2016) 779–790, https://doi.org/10.1016/j.jvcir.2016.07.011.
[24] G. Soria, L.M. Ortega Alvarado, F.R. Feito, Augmented and virtual reality for underground facilities management, J. Comput. Inf. Sci. Eng. 18 (4) (2018) 1–9, https://doi.org/10.1115/1.4040460.
[25] R. Napolitano, A. Blyth, B. Glisic, Virtual environments for visualizing structural health monitoring sensor networks, data, and metadata, Sensors (Basel) 18 (1) (2018) 1–14, https://doi.org/10.3390/s18010243.
[26] M.C. Li, S. Amerudin, Z. Mohamed Yusof, Development of augmented reality pipeline Visualiser (Arpv) application for visualising underground water pipeline, Int. Arch. Photogramm. Remote. Sens. Spat. Inf. Sci. XLII-4 (W16) (2019) 365–373, https://doi.org/10.5194/isprs-archives-XLII-4-W16-365-2019.
[27] G. Zhi, Z. Liao, W. Tian, X. Wang, J. Chen, A 3D dynamic visualization method coupled with an urban drainage model, J. Hydrol. 577 (2019) 1–14, https://doi.org/10.1016/j.jhydrol.2019.123988.
[28] A. Fenais, S.T. Ariaratnam, S.K. Ayer, N. Smilovsky, Integrating geographic information systems and augmented reality for mapping underground utilities, Infrastructures 4 (4) (2019) 1–17, https://doi.org/10.3390/infrastructures4040060.
[29] A. Fenais, S.T. Ariaratnam, N. Smilovsky, Assessing the accuracy of an outdoor augmented reality solution for mapping underground utilities, J. Pipeline Syst. Eng. Pract. 11 (3) (2020) 1–9, https://doi.org/10.1061/(ASCE)PS.1949-1204.0000474.
[30] X. Tan, L. Fan, Y. Huang, Y. Bao, Detection, visualization, quantification, and warning of pipe corrosion using distributed fiber optic sensors, Autom. Constr. 132 (2021) 1–13, https://doi.org/10.1016/j.autcon.2021.103953.
[31] X. Tan, A. Abu-Obeidah, Y. Bao, H. Nassif, W. Nasreddine, Measurement and visualization of strains and cracks in CFRP post-tensioned fiber reinforced concrete beams using distributed fiber optic sensors, Autom. Constr. 124 (2021) 1–13, https://doi.org/10.1016/j.autcon.2021.103604.
[32] D.K. O'Day, R. Weiss, S. Chiavari, D. Blair, Water Main Evaluation for Rehabilitation/Replacement, Denver, CO: American Water Works Association Research Foundation; Cincinnati, US-EPA, OH, 1987. https://nepis.epa.gov.
[33] B. Rajani, Y. Kleiner, Comprehensive review of structural deterioration of water mains: physically based models, Urban Water 3 (3) (2001) 151–164, https://doi.org/10.1016/S1462-0758(01)00032-2.
[34] M. Ahammed, R.E. Melchers, Probabilistic analysis of underground pipelines subject to combined stresses and corrosion, Eng. Struct. 19 (12) (1997) 988–994, https://doi.org/10.1016/S0141-0296(97)00043-6.
[35] AWWA, C101–67(R1977): ANSI/AWWA Standard for Thickness Design of Cast-iron Pipe, American Water Works Association, Denver, Colorado, 1997. https://www.awwa.org.
[36] B. Rajani, C. Zhan, S. Kuraoka, Pipe-soil interaction analysis of jointed water mains, Can. Geotech. J. 33 (3) (1996) 393–404, https://doi.org/10.1139/t96-061.
[37] M. Almahakeri, A. Fam, I.D. Moore, Experimental investigation of longitudinal bending of buried steel pipes pulled through dense sand, J. Pipeline Syst. Eng. Pract. 5 (2) (2014) 1–10, https://doi.org/10.1061/(ASCE)PS.1949-1204.0000141.
[38] Guidelines for the Seismic Design of Oil and Gas Pipeline Systems, American Society of Civil Engineers (ASCE), New York, 1984. ISBN: 978-0-87262-428-3.
[39] D.G. Honegger, R.W. Gailing, D.J. Nyman, Guidelines for the Seismic Design and Assessment of Natural Gas and Liquid Hydrocarbon Pipelines, 2002 4th International Pipeline Conference, Vol. 4th International Pipeline Conference, Parts A and B, 2002, pp. 563–570, https://doi.org/10.1115/IPC2002-27330.
[40] F. Ansari, Practical implementation of optical fiber sensors in civil structural health monitoring, J. Intell. Mater. Syst. Struct. 18 (8) (2007) 879–889, https://doi.org/10.1177/1045389X06075760.
[41] Z. Fang, H. Su, F. Ansari, Modal analysis of structures based on distributed measurement of dynamic strains with optical fibers, Mech. Syst. Signal Process. 159 (2021) 1–15, https://doi.org/10.1016/j.ymssp.2021.107835.
[42] Y. Yao, M. Yan, Y. Bao, Measurement of cable forces for automated monitoring of engineering structures using fiber optic sensors: a review, Autom. Constr. 126 (2021) 1–14, https://doi.org/10.1016/j.autcon.2021.103687.
[43] X. Tan, Y. Bao, Q. Zhang, H. Nassif, G. Chen, Strain transfer effect in distributed fiber optic sensors under an arbitrary field, Autom. Constr. 124 (2021) 1–13, https://doi.org/10.1016/j.autcon.2021.103597.
[44] B. Glisic, Comparative study of distributed sensors for strain monitoring of pipelines [J], Geotechn. Eng. (J. Southeast Asia Geotechn. Soc. SEAGS) 50 (2) (2019) 28–35. https://collaborate.princeton.edu/en/publications/comparative-study-of-distributed-sensors-for-strain-monitoring-of.
[45] D. Inaudi, B. Glisic, Long-range pipeline monitoring by distributed fiber optic sensing, J. Press. Vessel. Technol. 132 (1) (2009) 1–9, https://doi.org/10.1115/1.3062942.
[46] X. Feng, Y. Han, Z. Wang, H. Liu, Structural performance monitoring of buried pipelines using distributed fiber optic sensors, J. Civ. Struct. Heal. Monit. 8 (3) (2018) 509–516, https://doi.org/10.1007/s13349-018-0286-3.
[47] P. Ni, I.D. Moore, W.A. Take, Distributed fibre optic sensing of strains on buried full-scale PVC pipelines crossing a normal fault, Géotechnique 68 (1) (2018) 1–17, https://doi.org/10.1680/jgeot.16.P.161.
[48] H. Lan, G. Ma, N.A. Hoult, I.D. Moore, Stress concentrations due to simulated corrosion pits in buried metal pipes under longitudinal bending, J. Civ. Struct. Heal. Monit. 12 (4) (2022) 785–796, https://doi.org/10.1007/s13349-021-00522-8.
[49] M. Li, X. Feng, Multisensor data fusion-based structural health monitoring for buried metallic pipelines under complicated stress states, J. Civ. Struct. Heal. Monit. (2022) 1–13, https://doi.org/10.1007/s13349-022-00609-w.
[50] M. Nikles, B.H. Vogel, F. Briffod, S. Grosswig, F. Sauser, S. Luebbecke, A. Bals, T. Pfeiffer, Leakage detection using fiber optics distributed temperature monitoring, in: Vol. 5384, Proceedings of SPIE - The International Society for Optical Engineering, 2004, pp. 18–25, https://doi.org/10.1117/12.540270.
[51] M. Mahmoodian, C.Q. Li, Reliability-based service life prediction of corrosion-affected cast Iron pipes considering multifailure modes, J. Infrastruct. Syst. 24 (2) (2018) 1–8, https://doi.org/10.1061/(asce)is.1943-555x.0000417.
[52] A. Amirat, A. Mohamed-Chateauneuf, K. Chaoui, Reliability assessment of underground pipelines under the combined effect of active corrosion and residual stress, Int. J. Press. Vessel. Pip. 83 (2) (2006) 107–117, https://doi.org/10.1016/j.ijpvp.2005.11.004.
[53] R. Sadiq, B. Rajani, Y. Kleiner, Probabilistic risk analysis of corrosion associated failures in cast iron water mains, Reliab. Eng. Syst. Saf. 86 (1) (2004) 1–10, https://doi.org/10.1016/j.ress.2003.12.007.
[54] B. Rajani, A. Abdel-Akher, Re-assessment of resistance of cast iron pipes subjected to vertical loads and internal pressure, Eng. Struct. 45 (2012) 192–212, https://doi.org/10.1016/j.engstruct.2012.06.019.
[55] AIAA Digital Engineering Integration Committee, Digital Twin: Definition & Value, American Institute of Aeronautics and Astronautics (AIAA) and Aerospace Industries Association (AIA). https://www.aia-aerospace.org/report/digital-twin-paper, 2020.
[56] M.G. Kapteyn, J.V.R. Pretorius, K.E. Willcox, A probabilistic graphical model foundation for enabling predictive digital twins at scale, Nat. Computat. Sci. 1 (5) (2021) 337–347, https://doi.org/10.1038/s43588-021-00069-0.
[57] L. Ding, Y. Zhou, B. Akinci, Building information modeling (BIM) application framework: the process of expanding from 3D to computable nD, Autom. Constr. 46 (2014) 82–93, https://doi.org/10.1016/j.autcon.2014.04.009.
[58] P. Conejos, F. Martínez Alzamora, M. Hervás Carot, J. Alonso Campos, Development and Use of a Digital Twin for the Water Supply and Distribution Network of Valencia (Spain), 17th International Computing & Control for the Water Industry Conference (CCWI), Sept. 1, 2019-Sept. 4, 2019, pp. 2–4. Exeter, U. K, https://www.researchgate.net/publication/340236600_Development_and_Use_of_a_Digital_Twin_for_the_Water_Supply_and_Distribution_Network_of_Valencia_Spain.
[59] J.M.D. Delgado, L.J. Butler, I. Brilakis, M.Z.E.B. Elshafie, C.R. Middleton, Structural performance monitoring using a dynamic data-driven BIM environment, J. Comput. Civ. Eng. 32 (3) (2018) 1–17, https://doi.org/10.1061/(ASCE)CP.1943-5487.0000749.
[60] A. Rasheed, O. San, T. Kvamsdal, Digital twin: values, challenges and enablers from a modeling perspective, IEEE Access 8 (2020) 21980–22012, https://doi.org/10.1109/ACCESS.2020.2970143.
[61] S. Tesfamariam, B. Rajani, R. Sadiq, Possibilistic approach for consideration of uncertainties to estimate structural capacity of ageing cast iron water mains, Can. J. Civ. Eng. 33 (8) (2006) 1050–1064, https://doi.org/10.1139/l06-042.
[62] R. Amaya-Gómez, M. Sánchez-Silva, E. Bastidas-Arteaga, F. Schoefs, F. Muñoz, Reliability assessments of corroded pipelines based on internal pressure – a review, Eng. Fail. Anal. 98 (2019) 190–214, https://doi.org/10.1016/j.engfailanal.2019.01.064.
[63] T. Kurashima, M. Tateda, Thermal effects on the Brillouin frequency shift in jacketed optical silica fibers, Appl. Opt. 29 (15) (1990) 2219–2222, https://doi.org/10.1364/AO.29.002219.
[64] M. Xie, Z. Tian, A review on pipeline integrity management utilizing in-line inspection data, Eng. Fail. Anal. 92 (2018) 222–239, https://doi.org/10.1016/j.engfailanal.2018.05.010.
[65] V. Kucera, E. Mattsson, Atmospheric corrosion, in: F. Mansfeld (Ed.), Corrosion Mechanisms, Marcel Dekker, New York, 1987. ISBN: 978-1-003-06567-8.
[66] B.R. Ellingwood, Risk-informed condition assessment of civil infrastructure: state of practice and research issues, Struct. Infrastruct. Eng. 1 (1) (2005) 7–18, https://doi.org/10.1080/15732470412331289341.
[67] D.M. Frangopol, Life-cycle performance, management, and optimisation of structural systems under uncertainty: accomplishments and challenges, Struct. Infrastruct. Eng. 7 (6) (2011) 389–413, https://doi.org/10.1080/15732471003594427.
[68] M. Rausand, A. Barros, A. Hoyland, System Reliability Theory: Models, Statistical Methods, and Applications, 3rd edition, Wiley, New York, 1994. ISBN: 978–1–119-37395-7, https://www.wiley.com/en-us/System+Reliability+Theory%3A+Models%2C+Statistical+Methods%2C+and+Applications%2C+3rd+Edition-p-9781119373520.
[69] M. Soliman, D.M. Frangopol, Reliability and remaining life assessment of fatigue critical steel structures: integration of inspection and monitoring information, in:

Proceedings of Structures Congress 2013, May 2, 2013-May 4, 2013, Pittsburgh, PA. U.S, 2013, pp. 709–720, https://doi.org/10.1061/9780784412848.063.
[70] D.M. Liu, Frangopol, Utility and information analysis for optimum inspection of fatigue-sensitive structures, J. Struct. Eng. 145 (2) (2019) 1–12, https://doi.org/10.1061/(ASCE)ST.1943-541X.0002257.
[71] Y. Sahraoui, R. Khelif, A. Chateauneuf, Maintenance planning under imperfect inspections of corroded pipelines, Int. J. Press. Vessel. Pip. 104 (2013) 76–82, https://doi.org/10.1016/j.ijpvp.2013.01.009.
[72] H.-L. Chi, S.-C. Kang, X. Wang, Research trends and opportunities of augmented reality applications in architecture, engineering, and construction, Autom. Constr. 33 (2013) 116–122, https://doi.org/10.1016/j.autcon.2012.12.017.
[73] H.W. Chuah, Why and Who Will Adopt Extended Reality Technology? Literature Review, Synthesis, and Future Research Agenda, Available at SSRN, 2018, https://doi.org/10.2139/ssrn.3300469. Accessed date: December 13.
[74] G.A. Kumar, A.K. Patil, T.W. Kang, Y.H. Chai, Sensor fusion based pipeline inspection for the augmented reality system, Symmetry 11 (10) (2019) 1–19, https://doi.org/10.3390/sym11101325.
[75] E.M. Wetzel, W.Y. Thabet, The use of a BIM-based framework to support safe facility management processes, Autom. Constr. 60 (2015) 12–24, https://doi.org/10.1016/j.autcon.2015.09.004.
[76] X. Wang, P.E.D. Love, M.J. Kim, C.-S. Park, C.-P. Sing, L. Hou, A conceptual framework for integrating building information modeling with augmented reality, Autom. Constr. 34 (2013) 37–44, https://doi.org/10.1016/j.autcon.2012.10.012.
[77] V. Gomez-Jauregui, C. Manchado, J. Del-Castillo-Igareda, C. Otero, Quantitative evaluation of overlaying discrepancies in mobile augmented reality applications for AEC/FM, Adv. Eng. Softw. 127 (2019) 124–140, https://doi.org/10.1016/j.advengsoft.2018.11.002.
[78] Y. Deng, J.C.P. Cheng, C. Anumba, Mapping between BIM and 3D GIS in different levels of detail using schema mediation and instance comparison, Autom. Constr. 67 (2016) 1–21, https://doi.org/10.1016/j.autcon.2016.03.006.
[79] D.D.L. Mascareñas, J.P. Ballor, O.L. McClain, M.A. Mellor, C.-Y. Shen, B. Bleck, J. Morales, L.-M.R. Yeong, B. Narushof, P. Shelton, E. Martinez, Y. Yang, A. Cattaneo, T.A. Harden, F. Moreu, Augmented reality for next generation infrastructure inspections, Struct. Health Monit. 20 (4) (2020) 1957–1979, https://doi.org/10.1177/1475921720953846.
[80] D. Maharjan, M. Agüero, D. Mascarenas, R. Fierro, F. Moreu, Enabling human–infrastructure interfaces for inspection using augmented reality, Struct. Health Monit. 20 (4) (2020) 1980–1996, https://doi.org/10.1177/1475921720977017.
[81] M. Chu, J. Matthews, P.E.D. Love, Integrating mobile building information modelling and augmented reality systems: an experimental study, Autom. Constr. 85 (2018) 305–316, https://doi.org/10.1016/j.autcon.2017.10.032.
[82] R. Napolitano, Z. Liu, C. Sun, B. Glisic, Combination of image-based documentation and augmented reality for structural health monitoring and building pathology, Frontiers in built, Environment 5 (2019) 1–14, https://doi.org/10.3389/fbuil.2019.00050.
[83] T. Bailey, H. Durrant-Whyte, Simultaneous localization and mapping (SLAM): part II, IEEE Robot. Automat. Magaz. 13 (3) (2006) 108–117, https://doi.org/10.1109/MRA.2006.1678144.
[84] H. Bae, M. Golparvar-Fard, J. White, High-precision vision-based mobile augmented reality system for context-aware architectural, engineering, construction and facility management (AEC/FM) applications, Visualizat. Eng. 1 (1) (2013) 1–13, https://doi.org/10.1186/2213-7459-1-3.
[85] W. Li, Y. Han, Y. Liu, C. Zhu, Y. Ren, Y. Wang, G. Chen, Real-time location-based rendering of urban underground pipelines, ISPRS Int. J. Geo Inf. 7 (1) (2018) 1–17, https://doi.org/10.3390/ijgi7010032.
[86] P.X.W. Zou, P. Lun, D. Cipolla, S. Mohamed, Cloud-based safety information and communication system in infrastructure construction, Saf. Sci. 98 (2017) 50–69, https://doi.org/10.1016/j.ssci.2017.05.006.
[87] Boiler and Pressure Vessel Code, American Society of Mechanical Engineers (ASME), New York, 2012, https://doi.org/10.1115/1.859865.ch8. ISBN: 978-0-79185-986-5.
[88] Technical specification for directly buried hot-water heating pipeline in city, in: CJJ/T 81–2013, China, 2013, ISBN: 1511223799, 2013. Accessed July 26, 2013 from, http://www.mohurd.gov.cn/info/public.
[89] M. Staat, M. Heitzer, Numerical Methods for Limit and Shakedown Analysis: Deterministic and Probabilistic Problems, John von Neumann Institute for Computing, Central Institute for Applied Mathematics, 2003. ISBN: 3–00–010001-6, http://hdl.handle.net/2128/2926.
[90] General principles on reliability for structures, ISO2394. https://www.iso.org/obp/ui/#iso:std:iso:2394:ed-3:v1:en, 1998. (Accessed 16 October 2022).
[91] N.M. Okasha, D.M. Frangopol, A. Decò, Integration of structural health monitoring in life-cycle performance assessment of ship structures under uncertainty, Mar. Struct. 23 (3) (2010) 303–321, https://doi.org/10.1016/j.marstruc.2010.07.004.